

PROJECT REPORT

Privacy-Preserving Healthcare Device Authentication via Edge-AI and Federated Learning

Submitted to

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, KAKINADA

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BACHELOR OF TECHNOLOGY

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CERTIFICATE

This is to certify that this project report entitled “**Privacy-Preserving Healthcare Device Authentication via Edge-AI and Federated Learning**” is the Bonafide work of G.Vincy (22X41A4414), N.Yaswanth (22X41A4436), O.Nagendra (22X41A4439), K.Satyavathi (22X41A4418) , N.Vasu Deva Naga Sai (22X41A4437), SK.Yasin Ali (22X41A4453) in partial fulfillment of the requirements for the award of the graduate degree in BACHELOR OF TECHNOLOGY during the academic year 2025-2026. This Work has carried out under our supervision and guidance

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DECLARATION

We **G. Vincy, N. Yaswanth, O. Nagendra, K. Satyavathi, N. Vasu Deva Naga Sai, SK. Yasin Ali** declare that the project report entitled “**Privacy-Preserving Healthcare Device Authentication via Edge-AI and Federated Learning**” is an original work done in the Department of Computer Science & Engineering – Data Science, SRK Institute of Technology, Enikepadu, Vijayawada, during the academic year 2025-2026, in partial fulfillment for the award of the Degree of Bachelor of Technology in Computer Science & Engineering - Data Science. We assure you that this project is not submitted to any other College or University.

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ABSTRACT

Internet of Medical Things (IoMT) and hospital-edge devices to monitor patient health have accelerated the process of generating sensitive medical data. Although this type of data can make a significant contribution to the detection of the disease with the help of artificial intelligence (AI) but also traditional model faced several limitation such as data privacy, computational cost and over-fitting issues. To overcome the limitation this paper suggests a privacy-preserving Federated Learning (FL) model that incorporates a token-based dual-layer authentication and high-level medical image classification. The suggested system requires that hospitals will initially be registered in a Centralized Authorization Authority (CAA) where metadata of devices and hospital credentials are checked and HMAC-SHA256 tokens are generated. FL only accepts authenticated hospitals that possess active tokens, and rest of the nodes are not allowed, therefore safe and reliable participation is guaranteed. To train locally, each hospital applies Optimized Spatial Densely Connected Swin Transformer (OSDSTran), an architecture with a combination of spatial attention and transformer layers to successfully discern local and global characteristics of the chest X-ray images to classify the disease. The Prairie Dog Optimization Algorithm (PDOA) is then used to refine the hyper-parameters and to provide an accurate result. Authorized updates are fully secured by FedAvg, and the patient information is all local, which guarantees a high level privacy protection. The proposed method has been proven to be efficient in secure and reliable medical image classification with accuracy of 98.88% and precision of 98.65%, which will improve the model performance in real-time.

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

Medical devices connected via Internet are open to cyber-attacks, hence the privacy preservation of the user and the devices plays an important role. In this paper, Medical Edge-devices privacy enhanced technique based on the federated learning principles is proposed on the distributed coordination of devices and computational system. The increasing use of IoMT and hospital-edge devices for patient monitoring generates sensitive medical data that cannot be shared centrally due to privacy regulations, making traditional AI training risky. Standard FL ensures data locality but lacks robust authentication, leaving the global model vulnerable to malicious or unauthorized updates. Motivated by the need for secure, privacy-preserving, and trustworthy AI in healthcare, this work integrates token-based dual-layer authentication with federated learning to ensure only verified hospitals contribute to the global model while maintaining high model accuracy and patient data confidentiality.

In the proposed privacy-preserving FL framework, each hospital first registers with the centralized authorization authority (CAA), where devices and hospitals credentials (Hospital ID, device metadata) are recorded and **HMAC-SHA256** is issued for authentication. Active tokens are required, and only hospitals that are a part of the FL network are allowed to exclude the unauthorized or compromised nodes. After authentication, individual hospitals load its local medical dataset, that is, **Chest X ray** and local model training through **optimized densely connected Swin transformer based network** to classify the diseases. This method store the raw data locally to maintain patient privacy. The authentication mechanism verifies the hospital according to the data stored in **CAA**. Only validated updates are submitted to the central server to be securely aggregated, by **FedAvg**. Lastly, the authenticated local model updates are classified to improve privacy. Such a **token-based federated learning** is more favourable than the traditional FL, due to presence of **dual-layer authentication**, which stops the introduction of poisoned updates within the network to increase both the level of security and the strength of the model without sacrificing the level of privacy and adherence to the medical data policies.

1.2 ABOUT THE PROJECT

This project presents a **privacy-preserving healthcare framework** for secure medical data analysis using **Federated Learning (FL) and Edge-AI techniques**. The system is designed to address critical challenges in modern healthcare systems, such as **data privacy, unauthorized access, and high computational costs** associated with centralized machine learning models. By enabling distributed learning across multiple hospitals, the framework ensures that sensitive patient data remains local while still contributing to a globally optimized model.

The proposed system integrates a **token-based dual-layer authentication mechanism** with an advanced deep learning architecture for medical image classification. A **Centralized Authorization Authority (CAA)** is used to verify healthcare institutions and generate secure authentication tokens using **HMAC-SHA256**, ensuring that only trusted participants can contribute to the learning process. The framework further incorporates an optimized deep learning model, **OSDSTran (Optimized Spatial Densely Connected Swin Transformer)**, to extract both local and global features from medical images, improving classification accuracy.

To enhance model performance, a **Prairie Dog Optimization Algorithm (PDOA)** is applied for hyperparameter tuning, ensuring efficient convergence and improved accuracy. The system follows a federated learning approach where locally trained models are securely aggregated using the **FedAvg algorithm**, thereby maintaining data confidentiality while enabling collaborative learning. Overall, the framework provides a **secure, efficient, and scalable solution** for real-time healthcare applications with high accuracy and strong privacy protection.

The Algorithms Used in This Project Are Explained Below:

1.HMAC-SHA256 (Hash-Based Message Authentication Code)

This algorithm is used for secure authentication of healthcare devices and hospitals.

- Generates a unique authentication token using a secret key.
- Ensures data integrity and authenticity during communication.
- Prevents unauthorized access and malicious node participation.

2.OSDSTran (Optimized Spatial Densely Connected Swin Transformer)

This model is used for deep feature extraction and disease classification from medical images.

- Swin Transformer captures global dependencies using attention mechanisms.
- DenseNet connections improve feature reuse and reduce vanishing gradient issues.
- Spatial Attention Module (SAM) focuses on important regions in medical images.
- Produces high-quality feature representations for accurate classification.

. 3. PDOA (Prairie Dog Optimization Algorithm)

This is an optimization algorithm used for tuning model parameters.

- Automatically selects optimal hyperparameters.
- Improves model accuracy and convergence speed.
- Avoids local minima and enhances global search capability.

4. FedAvg (Federated Averaging Algorithm)

This algorithm is used for aggregating models in federated learning.

- Combines local model updates from multiple hospitals.
- Ensures data privacy by not sharing raw data.
- Reduces communication overhead and improves efficiency.
- Produces a robust global model through collaborative learning.

1.3 PURPOSE:

The main purpose of this project is to develop a secure, privacy-preserving, and efficient healthcare data analysis system using Federated Learning and Edge-AI techniques. With the rapid growth of IoMT devices, a large amount of sensitive patient data is generated, which requires strong protection against data breaches and unauthorized access.

The proposed system aims to overcome the limitations of traditional centralized machine learning methods by ensuring that patient data remains within local hospital environments while still enabling collaborative model training. Additionally, the project focuses on integrating robust authentication mechanisms and optimized deep learning models to enhance both security and accuracy in medical diagnosis.

The key purposes of this project are:

- To ensure **data privacy and security** by avoiding centralized data sharing.
- To implement a **token-based authentication mechanism (HMAC-SHA256)** for secure access.
- To develop an **accurate disease classification model (OSDSTran)** for medical images.
- To optimize model performance using **PDOA (Prairie Dog Optimization Algorithm)**.
- To enable **collaborative learning across multiple hospitals** using Federated Learning.
- To provide a **reliable and real-time healthcare decision support system**.

1.4 SCOPE:

The scope of this study covers the functional, technical, and application aspects of the proposed privacy-preserving healthcare framework using Federated Learning and Edge-AI techniques. The system is designed to operate on medical image datasets (such as chest X-rays) and focuses on secure disease classification while ensuring patient data privacy. By integrating authentication mechanisms, deep learning models, and optimization techniques, the framework ensures high accuracy, security, and scalability in healthcare environments.

1. Functional Scope

- **Authentication and Security:**

The system uses **HMAC-SHA256-based token authentication** through a Centralized Authorization Authority (CAA).

- **Local Model Training:**

Each hospital trains the model locally using its own medical dataset.

This ensures that **sensitive patient data remains within the hospital**, maintaining privacy and complying with data protection policies.

- **Feature Extraction and Classification:**

The system utilizes the **OSDSTran (Optimized Spatial Densely Connected Swin Transformer)** model.

- Captures **local features** (texture, edges, abnormalities) using DenseNet
- Captures **global dependencies** using Swin Transformer

Applies **spatial attention** to focus on important regions

- **Model Optimization:**

The **Prairie Dog Optimization Algorithm (PDOA)** is used to optimize hyperparameters.

- Improves model accuracy
- Reduces training time
- Enhances convergence efficiency

2. Technical Scope

- **Dataset:**

The system is trained and tested using medical image datasets (Chest X-ray dataset) for disease detection. The dataset contains labeled images representing different medical conditions, enabling effective supervised learning.

- **Pre-processing Techniques:**

The input images undergo preprocessing to improve quality and consistency:

- Image resizing and normalization
- Noise reduction techniques
- Data augmentation (if applied)
- Scaling pixel values for uniformity

- **Model Architecture:**

The system integrates multiple advanced techniques:

- OSDSTran model for feature extraction and classification
- PDOA optimization for parameter tuning
- Federated Learning framework for decentralized training
- HMAC-SHA256 authentication for secure access

3. Application Scope

- The system can be applied in hospitals, diagnostic centers, and IoMT-based healthcare environments for secure and accurate disease detection, supporting doctors in decision-making while preserving patient data privacy.

4. Future Scope

- The system can be enhanced by integrating blockchain for security, extending to multi-class disease detection, supporting additional datasets like MRI and CT scans, and deploying as a web or mobile-based real-time healthcare application.

CHAPTER 2

LITERATURE REVIEW

Paper 1: Ahmed et al. – IEEE Transactions on Consumer Electronics (2025)

Title: Privacy Enhanced Edge-AI Healthcare Devices Authentication: A Federated Learning Approach

Detailed Explanation:

- The study focuses on privacy preservation in IoMT (Internet of Medical Things) using Federated Learning.
- Introduces a Centralized Authorization Authority (CAA) for managing device authentication.
- Uses a token-based dual-layer authentication mechanism:
 - Authentication Authority Validation Layer (AAVL)
 - Centralized Server Firewall Authentication Layer (CSFAL)
- Ensures that only authorized healthcare devices participate in model training.
- Data remains local to devices, reducing privacy risks.

Models/Techniques Used:

- Federated Learning (FL)
- Token-based authentication
- Edge-AI framework

Key Contribution:

- Combines **privacy preservation with authentication** in IoMT systems
- Prevents **unauthorized access and malicious attacks**
- Improves **security in distributed healthcare environments**

Results:

- Achieved approximately **96% accuracy**
- Demonstrated improved **privacy and system reliability**

Paper 2: Ayyappan et al. – International Conference on Self Sustainable AI Systems (ICSSAS, 2025)

Title: Lightweight Architecture for IoT-Based Keyword Spotting

Detailed Explanation:

- Focuses on **efficient deep learning models for IoT devices**.
- Uses lightweight neural architectures to reduce **computational cost and latency**.
- Emphasizes **real-time processing** in resource-constrained environments.

Models/Techniques Used:

- Convolutional Neural Networks (CNN)
- Lightweight deep learning architectures

Key Contribution:

- Reduces **memory usage and computation time**
- Suitable for **edge devices and IoT systems**

Results:

- Achieved **efficient performance with reduced latency**
- Demonstrated suitability for **real-time applications**

Paper 3: Yang et al. – Healthcare Analytics (2025)

Title: P2DCA: Privacy-Preserving Data Collection and Analysis Framework for IoMT

Detailed Explanation:

- Proposes a secure data collection framework for IoMT applications.
- Focuses on privacy-preserving data transmission and storage.
- Uses distributed architecture to ensure secure communication.

Models/Techniques Used:

- Privacy-preserving algorithms
- Distributed data processing

Key Contribution:

- Enhances **data security during transmission**
- Reduces risk of **data leakage in healthcare systems**

Results:

- Improved **data confidentiality and integrity**
- Suitable for **large-scale IoMT systems**

Paper 4: Eshwarappa et al. – Journal of Cloud Computing (2025)

Title: Privacy Preservation in IoMT Using Federated Learning and Blockchain

Detailed Explanation:

- Combines **Federated Learning with Blockchain technology**.
- Ensures **secure and transparent data sharing** in healthcare systems.

Models/Techniques Used:

- Federated Learning
- Blockchain technology

Key Contribution:

- Provides **high security and transparency**
- Eliminates dependency on centralized systems

Results:

- Improved **data trust and system security**
- Suitable for **secure healthcare applications**

Paper 5: Chen et al. – Discover Applied Sciences (2025)

Title: Federated Reinforcement-Based Fusion Model for IoMT Security

Detailed Explanation:

- Proposes a **federated reinforcement learning approach** for IoMT security.
- Focuses on detecting **cyber-attacks and malicious activities**.
- Uses adaptive learning to improve system performance over time.

Models/Techniques Used:

- Federated Learning
- Reinforcement Learning

Key Contribution:

- Enhances **security against cyber threats**
- Improves **adaptive learning capability**

Results:

- Achieved better **attack detection accuracy**
- Improved **system robustness**-specific modeling.

CHAPTER - 3

PROPOSED METHODOLOGY

3.1 EXISTING SYSTEM

Existing healthcare data analysis systems mainly rely on **centralized machine learning and deep learning approaches** for disease detection using medical datasets such as chest X-rays. In these systems, data from multiple hospitals is collected and stored in a central server for training models. Traditional machine learning techniques depend on **manual feature extraction**, which limits the model's ability to capture complex patterns in medical images.

Advanced deep learning models such as **Convolutional Neural Networks (CNN), ResNet, and Vision Transformers (ViT)** have improved feature extraction and classification accuracy. However, these approaches require **large-scale centralized datasets**, leading to serious concerns regarding **data privacy, security, and unauthorized access**. Additionally, existing systems lack strong authentication mechanisms, making them vulnerable to **malicious attacks and data leakage**.

Although Federated Learning has been introduced to address privacy issues by keeping data local, many existing FL-based systems still suffer from **weak authentication, inefficient optimization, and limited model performance**. Overall, these limitations affect the **reliability, scalability, and practical deployment** of healthcare AI systems.

3.1.1 DISADVANTAGES OF EXISTING SYSTEM

- Centralized storage of patient data increases the risk of data breaches and privacy issues.
- Weak authentication mechanisms allow unauthorized access to the system.
- Dependence on centralized architecture reduces security and scalability.
- High computational requirements make deployment difficult in real-time environments.
- Large labeled datasets are required, which are hard to obtain due to privacy constraints.
- Manual feature extraction limits the model's ability to capture complex patterns.
- Inefficient handling of distributed data leads to delays and reduced performance.
- Systems are vulnerable to cyber-attacks, data tampering, and model poisoning.
- Lack of optimization techniques results in lower accuracy and slower convergence.
- Overall system reliability is reduced for real-time healthcare applications.

3.2 PROPOSED SYSTEM

The proposed system introduces a **secure and privacy-preserving healthcare framework** using **Federated Learning, Edge-AI, and advanced deep learning techniques**. The system ensures that sensitive patient data remains within local hospital environments while enabling collaborative model training across multiple institutions.

A **Centralized Authorization Authority (CAA)** is used to register hospitals and generate secure authentication tokens using **HMAC-SHA256**, ensuring that only authorized devices can participate in the learning process. The system employs an advanced deep learning model, **OSDSTran (Optimized Spatial Densely Connected Swin Transformer)**, for accurate disease classification from medical images.

To enhance performance, the **Prairie Dog Optimization Algorithm (PDOA)** is applied for hyperparameter tuning, improving accuracy and reducing training complexity. The federated learning process aggregates local model updates using the **FedAvg algorithm**, ensuring data privacy and efficient collaboration.

The working of the proposed system is carried out in different stages as explained below:

- Collects medical image data (e.g., chest X-rays) from multiple hospitals
- Registers hospitals through **CAA and generates authentication tokens**
- Pre-processes images using normalization and resizing techniques
- Trains local models using **OSDSTran architecture**
- Extracts both **local and global features** using DenseNet and Swin Transformer
- Applies **spatial attention mechanisms** to focus on important regions
- Optimizes model parameters using **PDOA algorithm**
- Shares only model updates (not raw data) to the central server
- Aggregates models using **FedAvg algorithm**
- Updates the global model and redistributes it to hospitals
- Produces accurate disease classification results
- Ensures secure and privacy-preserving healthcare data processing

3.2.1 Proposed Methodology:

- ▶ **Data Collection:**
Collect medical image datasets (such as chest X-rays) from multiple healthcare institution
- ▶ **Authentication:**
Register hospitals using CAA and generate secure tokens using HMAC-SHA256.
- ▶ **Pre-processing:**
Apply image resizing, normalization, and noise reduction techniques.
- ▶ **Feature Extraction:**
Extract features using the OSDSTran model combining Swin Transformer and DenseNet.
- ▶ **Attention Mechanism:**
Apply spatial attention to highlight important regions in medical images.
- ▶ **Local Model Training:**
Train models locally within hospitals without sharing raw data.
- ▶ **Optimization:**
Use PDOA algorithm to tune hyperparameters and improve performance.
- ▶ **Model Aggregation:**
Combine local model updates using FedAvg in the federated learning framework.
- ▶ **Global Model Update:**
Update and distribute the global model to all participating hospitals.
- ▶ **Output Generation:**
Produce accurate and secure disease classification results for healthcare applications.

SYSTEM ARCHITECTURE

The proposed system architecture is designed to provide a secure, privacy-preserving, and efficient healthcare data analysis framework using Federated Learning and Edge-AI techniques. The architecture follows a decentralized approach, where multiple hospitals (clients) collaboratively train a global model without sharing sensitive patient data.

The system consists of three main components:

Local Hospitals (Clients), Centralized Authorization Authority (CAA), and Global Server (Federated Aggregation Layer).

Architecture Components :

1. Local Hospitals (Client Layer)

- Each hospital collects **medical image data (e.g., chest X-rays)**.
- Data is **preprocessed locally** (normalization, resizing).
- The **OSDSTran model** is trained on local data.
- Extracts **local + global features** using DenseNet and Swin Transformer.
- Only **model updates (weights)** are sent to the central server.
- Raw data is never shared → ensures **privacy**

2. Centralized Authorization Authority (CAA)

- Responsible for **registration of hospitals and devices**.
- Generates secure tokens using **HMAC-SHA256**.
- Verifies identity of participants before allowing access.
- Prevents **unauthorized or malicious nodes** from joining.
- Ensures **secure communication** between clients and server.

3. Global Server (Federated Learning Layer)

- Receives model updates from multiple hospitals.
- Aggregates updates using **FedAvg algorithm**.
- Creates an improved **global model**.
- Sends updated model back to all hospitals.
- No access to raw patient data → maintains **confidentiality**.

4. Optimization Module (PDOA)

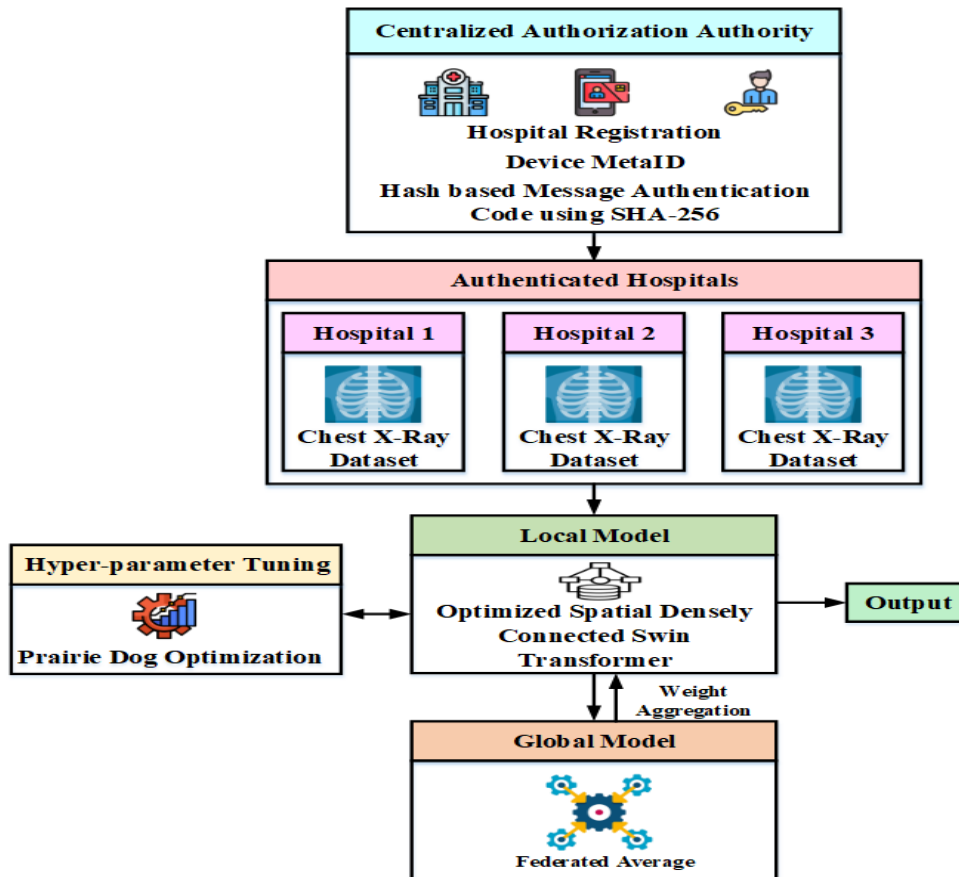
- Tunes hyperparameters of the model.
- Improves accuracy and convergence speed.
- Reduces computational complexity.

Working Flow of Architecture

1. Hospitals register with CAA and receive authentication tokens
2. Medical data is collected and processed locally
3. Local models are trained using OSDSTran
4. Features are extracted and optimized using PDOA
5. Only model updates are sent to the global server

6. Server aggregates models using FedAvg
7. Updated global model is shared back to hospitals

Process repeats until optimal performance is achieved



3.2.2. ADVANTAGES OF PROPOSED SYSTEM

1. Enhanced Data Privacy

The system uses Federated Learning, where patient data remains within local hospitals. Only model updates are shared, ensuring high privacy and protection of sensitive information.

2. Secure Authentication Mechanism

The integration of HMAC-SHA256 with Centralized Authorization Authority (CAA) ensures that only authorized hospitals and devices can access the system, preventing unauthorized participation

3. Decentralized Learning Framework

The proposed system eliminates centralized data storage, making it more secure, scalable, and suitable for distributed healthcare environments.

4. Improved Feature Extraction

The OSDSTran model effectively captures both local and global features from medical images, improving the quality of feature representation.

5. High Accuracy and Reliability

By combining advanced deep learning and optimization techniques, the system achieves high accuracy (around 98%+), making it reliable for medical diagnosis.

6. Efficient Model Optimization

The Prairie Dog Optimization Algorithm (PDOA) automatically tunes hyperparameters, improving model performance and convergence speed.

3.3 HARDWARE REQUIREMENTS

The proposed system requires the following hardware components for efficient training and execution of the model:

- **Processor:** Intel Core i5 / i7 (or higher) Required for handling data processing and model training tasks
- **RAM:** Minimum 8 GB (16 GB recommended) Supports smooth execution of deep learning and federated learning operations
- **Storage:** Minimum 256 GB HDD/SSD (SSD preferred) Used for storing datasets, models, and outputs
- **GPU (Optional but Recommended):** NVIDIA GPU (e.g., RTX 3050/3060) Accelerates deep learning model training and reduces computation time
- **System Architecture:** 64-bit system Required for compatibility with modern ML/DL frameworks

3.4 SOFTWARE REQUIREMENTS

The proposed system is implemented using the following software tools and technologies:

- **Operating System:** Windows 10/11 or Linux (Ubuntu preferred) Provides environment for development and execution
- **Programming Language:** Python Used for implementing machine learning, deep learning, and federated learning models
- **Libraries/Frameworks:**
 - **TensorFlow / Keras:** For building and training deep learning models
 - **NumPy & Pandas:** For data manipulation and preprocessing
 - **Matplotlib / Seaborn:** For data visualization
 - **Scikit-learn:** For evaluation metrics and preprocessing
- **Development Tools:**
 - Jupyter Notebook / Google Colab / VS Code
 - Used for coding, testing, and experimentation
- **Database (Optional):** MySQL or local file storage
 - For storing datasets and results

Other Tools: OpenCV (for image processing if required) ,HMAC & hashlib libraries (for authentication implementation)

CHAPTER 4

SYSTEM DESIGN

4.1 DATA DICTIONARY

	Image_Name	Label
0	NORMAL2-IM-1221-0001.jpeg	NORMAL
1	NORMAL2-IM-1222-0001.jpeg	NORMAL
2	NORMAL2-IM-1223-0001.jpeg	NORMAL
3	NORMAL2-IM-1224-0001.jpeg	NORMAL
4	NORMAL2-IM-1225-0001.jpeg	NORMAL
5	NORMAL2-IM-1226-0001.jpeg	NORMAL
6	NORMAL2-IM-1227-0001.jpeg	NORMAL
7	NORMAL2-IM-1228-0001.jpeg	NORMAL
8	NORMAL2-IM-1231-0001.jpeg	NORMAL
9	NORMAL2-IM-1232-0001.jpeg	NORMAL

FIG 2: Sample Dataset and its Attributes

DATA SET ATTRIBUTES:

To train a deep learning model for disease detection using Federated Learning, the dataset should include medical image data, patient metadata, preprocessing details, model information, and evaluation metrics. These attributes help in accurate classification while ensuring privacy and security.

FEATURES

Data	Attributes	Description
Patient Data	Patient_ID, Age, Gender	Stores basic demographic details of the patient
MedicalImage Data	Image_ID, Image_Path, Label	Contains chest X-ray images labeled as Normal or Pneumonia
DatasetInformation	Dataset_Name, Source	Information about dataset (e.g., Kaggle, hospital source)

Preprocessed Data	Resized_Image, Normalized_Image	Images after preprocessing like resizing and normalization
Feature Extraction	Local_Features, Global_Features	Features extracted using OSDSTran (DenseNet + Swin Transformer)
Attention Features	Spatial_Attention_Map	Highlights important regions in medical images
Model Information	Model_ID, Model_Name, Model_Type	Details of deep learning model used (OSDSTran)
Federated Learning Info	Client_ID, Local_Model_Update	Information about participating hospitals and their updates
Authentication Data	Token_ID, HMAC_Key	Security credentials used for validating authorized clients
Prediction Data	Prediction_ID, Predicted_Class	Output classification (Normal / Pneumonia)

TABLE NO. 1: Data Dictionary

DATA TYPES:

Data types play a crucial role in this project because selecting appropriate data types ensures:

- **Accurate data preprocessing** - of medical images and metadata
- **Efficient model training** - in deep learning frameworks
- **Proper feature extraction** - and representation
- **Reduced memory** - usage and faster computation
- **Fewer runtime-** and type-related errors
- **Improved** - overall model performance and reliability

LIBRARIES USED:

- NumPy
- Pandas
- Matplotlib
- Data time, Scikit-learn

4.2 ER DIAGRAM

The **Entity Relationship (ER) Diagram** represents the structural design of the proposed healthcare system. It illustrates the **entities, attributes, and relationships** involved in the system for efficient data organization and management.

The ER diagram helps in understanding how different components such as patient data, medical images, model processing, and federated learning modules are interconnected. It provides a clear visualization of data flow and interactions between various system elements.

Entities Description

- **Patient:**
Stores basic patient information such as Patient_ID, age, and gender.
- **Medical Data:**
Contains medical images (e.g., chest X-rays) along with labels indicating whether the condition is normal or diseased.
- **Preprocessing:**
Represents the preprocessing stage where images are resized and normalized for model training.
- **Model:**
Includes details of the deep learning model used for classification (OSDSTran).
- **Prediction:**
Stores the output results generated by the model, including predicted class and confidence score.
- **Hospital (Client):**
Represents different hospitals participating in the federated learning process.
- **Federated Update:**
Stores model updates generated by each hospital during training.
- **Authentication:**
Maintains security credentials such as tokens and keys for validating authorized users.

Relationships

- A Patient can have multiple Medical Data records (1:M)
- Each Medical Data undergoes Preprocessing (1:1)
- Medical Data is used for generating Predictions (1:M)
- A Model produces multiple Predictions (1:M)
- A Hospital generates multiple Federated Updates (1:M)
- Each Hospital has one Authentication record (1:1)

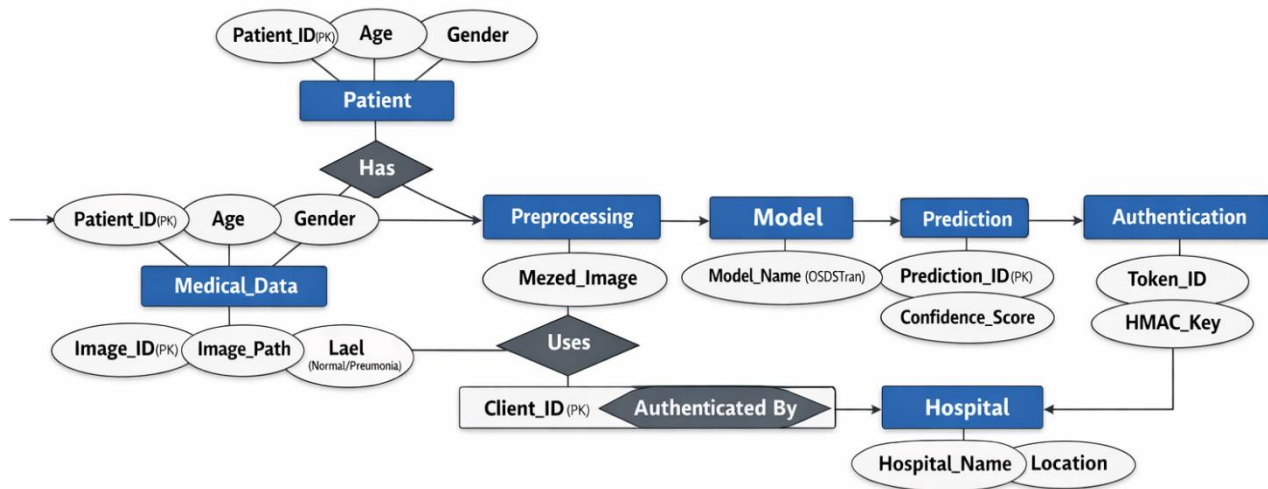


FIG 3: Representation of ER Diagram

4.3 DFDs DIAGRAM

The Data Flow Diagram (DFD) represents how data moves in your system:

“Privacy-Preserving Healthcare Device Authentication using Federated Learning”

It shows:

- How hospitals/devices send data
- How authentication is done
- How models are trained and aggregated securely

Components of DFD

A Data Flow Diagram has 4 main components:

1. External Entity

- Source or destination of data
- Example (your project)

2. Process

- Converts input data into output
- Represented by a circle/oval
- Example: Authentication, Model Training

3. Data Flow

- Shows movement of data
- Represented by arrows
- Example: Medical data → Server, Model updates → Global model

4. Data Store

- Stores data
- Represented by open rectangle

TYPES OF DFD's:

➤ Logical DFD

A Logical Data Flow Diagram represents **what the system does** by showing the flow of data between processes without including technical or implementation details.

➤ Physical DFD

A Physical Data Flow Diagram represents **how the system is implemented** by showing actual components like hardware, software, databases, and people involved in the system.

➤ Level-0 DFD (Context Diagram)

Level-0 DFD shows the **entire system as a single process** and highlights the interaction between the system and external entities.

➤ Level-1 DFD

Level-1 DFD breaks down the main system into **major internal processes** and shows how data flows between them.

4.3.1 DFDs Level-0 DIAGRAM

The Level 0 DFD represents the entire system as a single process where hospitals or IoMT devices interact with the Federated Learning system. Hospitals provide medical data and authentication credentials, and in return, receive predictions and updated global models.

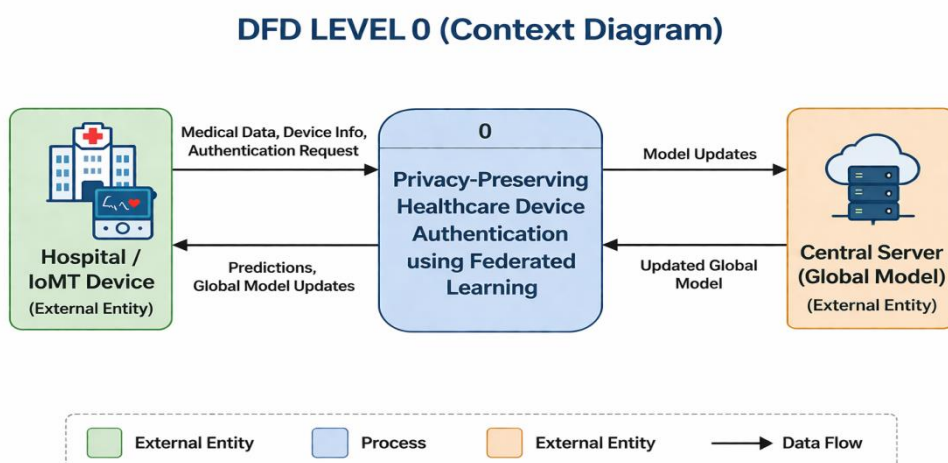


FIG 4: Representation of DFD Level-0 Diagram

4.3.2 DFDs Level-1 DIAGRAM

The Level 1 DFD decomposes the system into key processes such as device registration, authentication, local model training, and global model aggregation. Hospitals first register with the Centralized Authorization Authority (CAA), where authentication tokens are generated using HMAC-SHA256. Only authenticated devices are allowed to participate in federated learning. Each hospital trains the model locally using its dataset and sends only model updates to the central server, ensuring data privacy. The global server aggregates these updates using the FedAvg algorithm and sends the improved model back to hospitals.

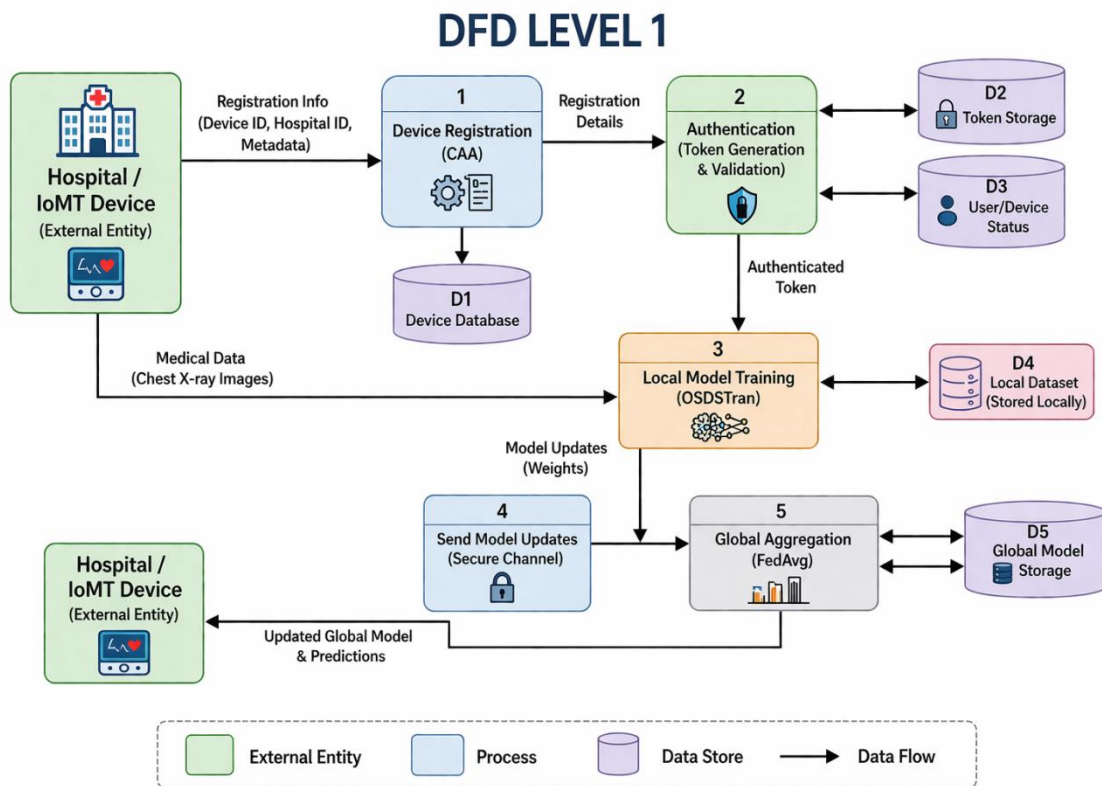


FIG 5: Representation of DFD Level-1 Diagram

4.4 UML DIAGRAMS

- **Unified Modeling Language (UML)** is a standard visual language used to design, visualize, and document software systems. It helps in representing both the structure and behavior of a system in a clear and understandable way.
- UML diagrams provide a graphical representation of system components such as classes, objects, processes, and their interactions. They make complex systems easier to understand for developers, designers, and stakeholders.
- In this project, UML diagrams are used to represent how hospitals, devices, and servers interact, how authentication is performed, and how federated learning processes are executed securely.

GOALS

The Primary goals in the design of the UML are as follows:

- Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
- Provide extendibility and specialization mechanisms to extend the core concepts.
- Be independent of particular programming languages and development process.
- Provide a formal basis for understanding the modeling language.
- Encourage the growth of OO tools market.
- Support higher level development concepts such as collaboration, patterns and components.
- Integrate best practices.

4.4.1 USE CASE DIAGRAM

The Use Case Diagram represents the interaction between users and the system in a simple and clear way. It identifies actors such as hospitals or IoMT devices and shows the functions they perform like registration, authentication, and model training. This diagram helps in understanding what the system does from the user's perspective. It also ensures that all required functionalities are included. Overall, it provides a high-level view of system behavior.

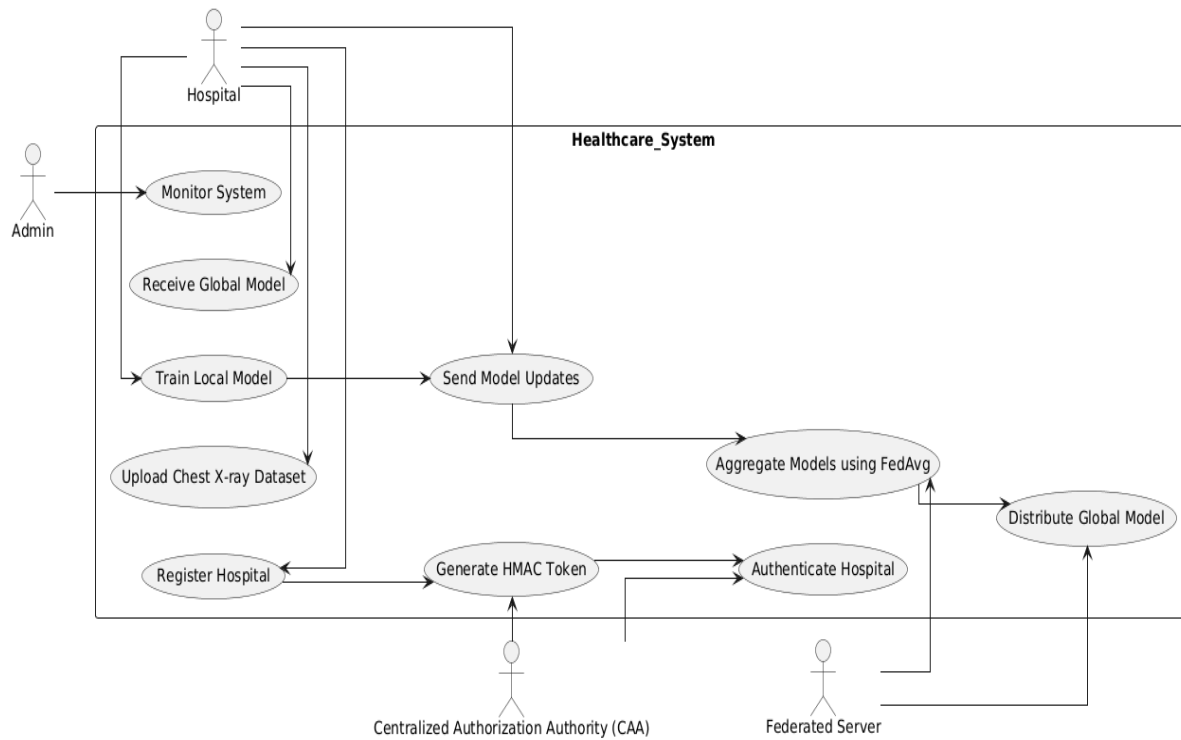


FIG 6: Representation of Use Case Diagram

4.4.2. CLASS DIAGRAM

The Class Diagram shows the structural design of the system by representing different classes, their attributes, and methods. It explains how various components like devices, tokens, and models are organized within the system. Relationships such as association and dependency between classes are also illustrated. This diagram acts as a blueprint for coding and implementation. It helps developers understand how the system is built internally.

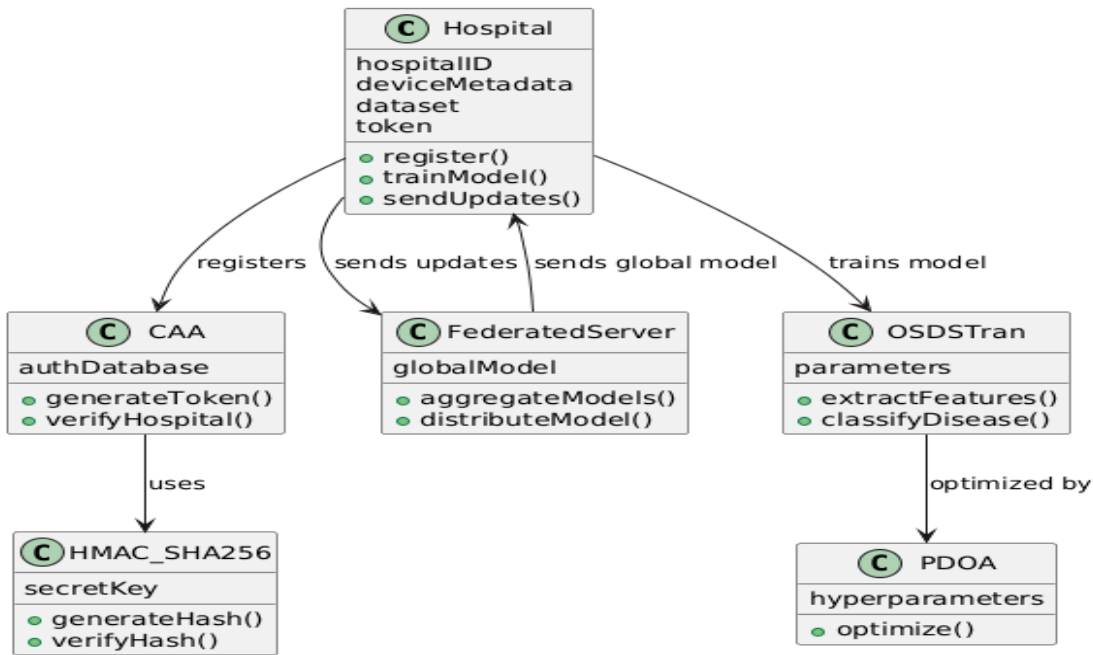
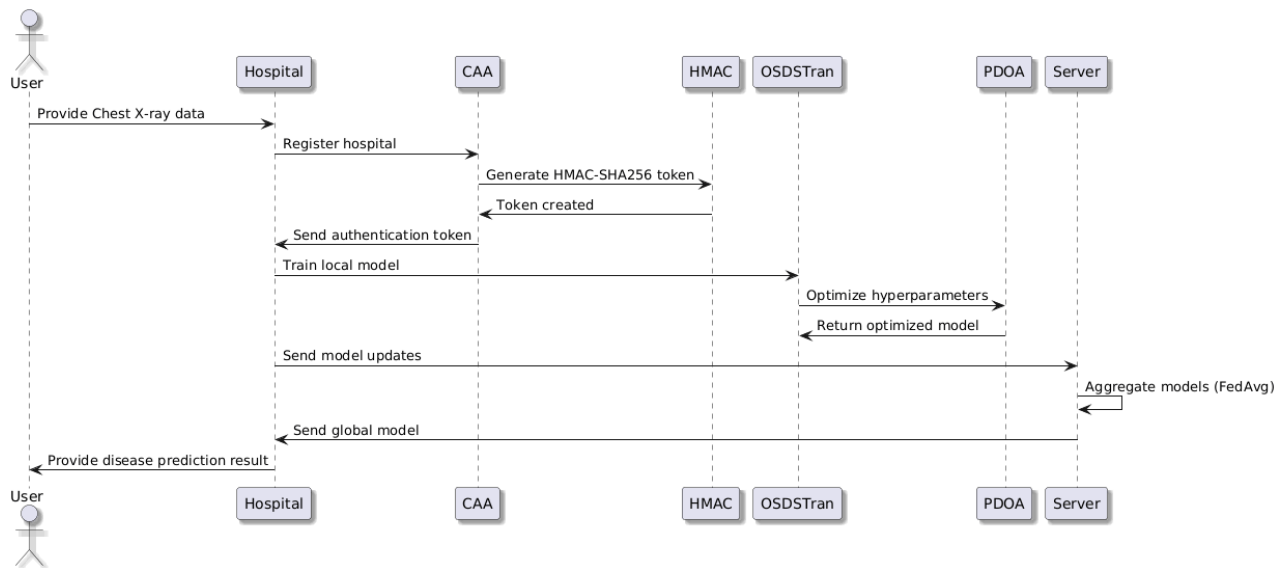


FIG 7: Representation of Class Diagram

4.4.3 SEQUENCE DIAGRAM

The Sequence Diagram describes how different components interact with each other in a sequential manner over time. It shows the order of operations such as device registration, authentication, local training, and sending model updates. Messages are passed between different entities in a step-by-step process. This helps in understanding the communication flow clearly. It is very useful for representing Real-time system behaviour

Fig No 8: REPRESENTATION OF SEQUENCE DIAGRAM

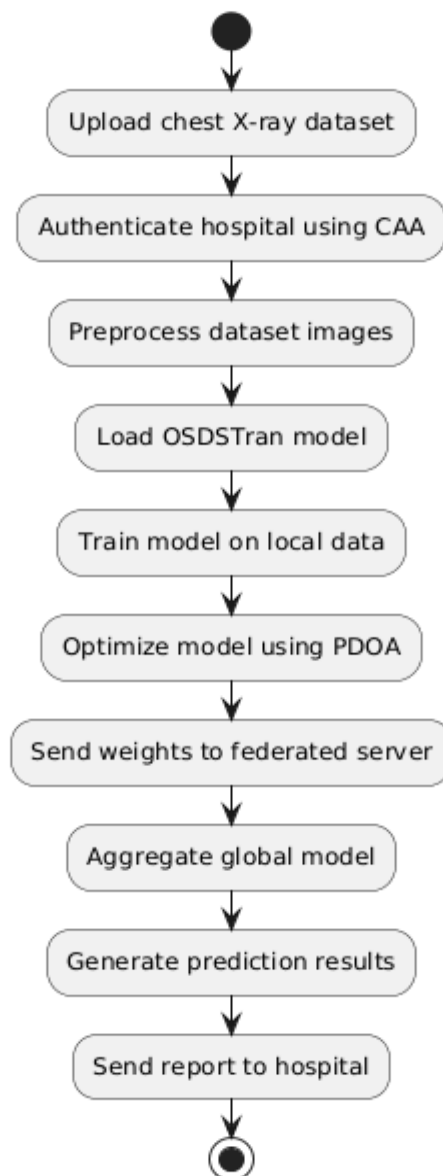


4.4.4 ACTIVITY DIAGRAM

The Activity Diagram represents the workflow of the system using flowchart-like symbols. It shows the sequence of activities such as authentication, training, and aggregation along with decision-making steps. It helps in understanding how the system processes data from start to end. Conditional flows like success or failure of authentication are also included. This diagram simplifies complex processes into easy visual steps.

Fig No 9 : REPRESENTATION OF ACTIVITY DIAGRAM

Activity Diagram - Healthcare Federated Learning Workflow



4.4.5 COLLABORATION DIAGRAM

The Collaboration Diagram represents how different components of the system interact with each other to achieve a specific task. It focuses on the relationships between objects such as hospital, authentication system, and central server. In this diagram, messages are exchanged between components in a numbered sequence to show the flow of communication. It helps in understanding how data moves between different parts of the system during processes like authentication and model training. Overall, it provides a clear view of object interactions and coordination within the system.

Collaboration Diagram - Healthcare Federated Learning System

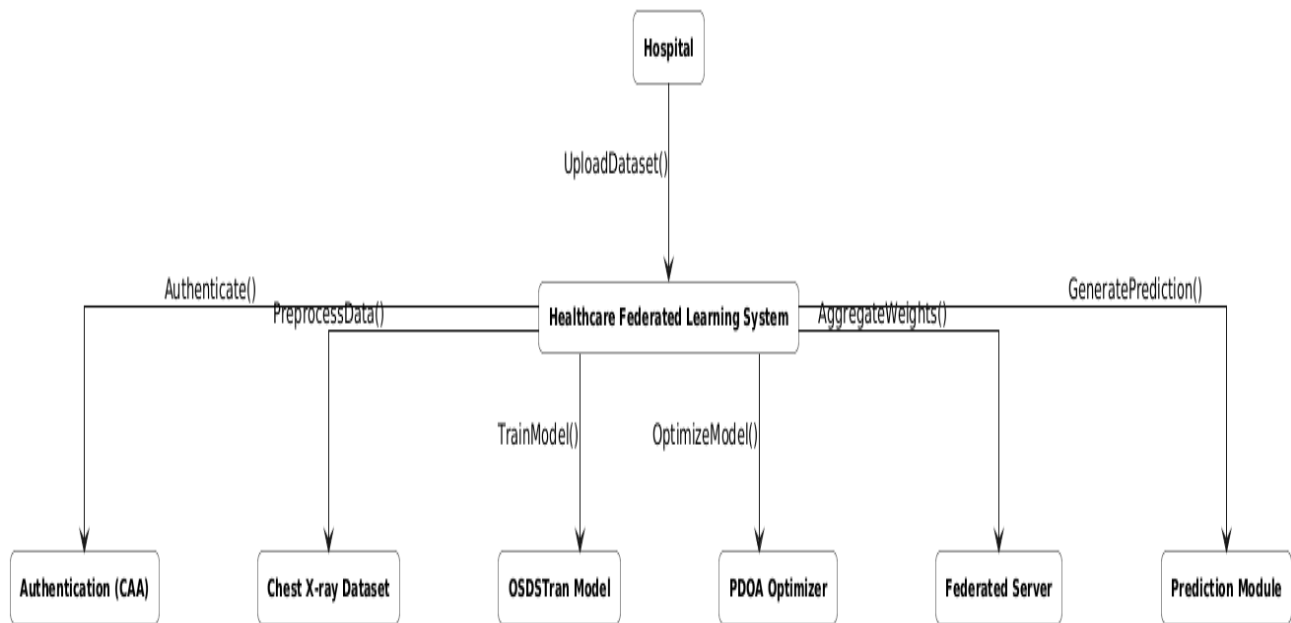


Fig No 10 : REPRESENTATION COLLABORATION D

CHAPTER 5

SYSTEM IMPLEMENTATION

INSTALL REQUIRED LIBRARIES:

```
!pip install tensorflow opencv-python matplotlib scikit-learn
```

IMPORT LIBRARIES

```
import numpy as np
import cv2
import os
import tensorflow as tf
import matplotlib.pyplot as plt
import hmac
import hashlib
import pickle
from sklearn.model_selection import train_test_split
from sklearn.metrics import precision_score, recall_score, f1_score
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
```

DEFINE HMAC FUNCTIONS

```
SECRET_KEY = b"secure_shared_key"
def generate_hmac(data, key):
    message = pickle.dumps(data)
    return hmac.new(key, message, hashlib.sha256).hexdigest()
    def verify_hmac(data, received_hmac, key):
        expected = generate_hmac(data, key)
        return hmac.compare_digest(expected, received_hmac)
```

LOAD DATASET

```
base = "dataset"
IMG_SIZE = 128
def load_data():
    data, labels = [], []
```

```

for label, cls in enumerate(["NORMAL", "PNEUMONIA"]):
    folder = os.path.join(base, cls)
    for img_name in os.listdir(folder)[:300]:
        img_path = os.path.join(folder, img_name)
        img = cv2.imread(img_path)
        if img is None:
            continue
        img = cv2.resize(img, (IMG_SIZE, IMG_SIZE)) / 255.0
        data.append(img)
        labels.append(label)
    return np.array(data), np.array(labels)
X, y = load_data()
print("Dataset:", X.shape)

```

SPLIT DATA

a) Train/Test Split

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

b) Client Split (for Federated Learning)

```
X1, X_temp, y1, y_temp = train_test_split(X_train, y_train, test_size=0.66)
```

```
X2, X3, y2, y3 = train_test_split(X_temp, y_temp, test_size=0.5)
```

```
clients = [(X1, y1), (X2, y2), (X3, y3)]
```

DATA AGUMENTATION

```

datagen = tf.keras.preprocessing.image.ImageDataGenerator(
    rotation_range=10,
    zoom_range=0.1,
    horizontal_flip=True
)

```

MODEL CREATION

```

def create_model():
    model = tf.keras.Sequential([
        tf.keras.layers.Conv2D(32, (3,3), activation='relu', input_shape=(128,128,3)),

```

```

tf.keras.layers.MaxPooling2D(),
tf.keras.layers.Conv2D(64, (3,3), activation='relu'),
tf.keras.layers.MaxPooling2D(),
tf.keras.layers.Conv2D(128, (3,3), activation='relu'),
tf.keras.layers.MaxPooling2D(),

tf.keras.layers.Flatten(),
tf.keras.layers.Dense(128, activation='relu'),
tf.keras.layers.Dropout(0.5),
tf.keras.layers.Dense(1, activation='sigmoid')
])
model.compile(optimizer='adam',
              loss='binary_crossentropy',
              metrics=['accuracy'])
return model

```

Federated Averaging Function

```

def federated_avg(weights):
    avg = [ ]
    for layer in zip(*weights):
        avg.append(np.mean(layer, axis=0))
    return avg

```

Federated Training

```

global_model = create_model()
history = [ ]
ROUNDS = 3
for r in range(ROUNDS):
    print(f"\n--- Round {r+1} ---")
    local_weights = [ ]
    for i in range(3):

        model = create_model()

```



```

model.set_weights(global_model.get_weights())
X_local, y_local = clients[i]
model.fit(datagen.flow(X_local, y_local, batch_size=16),
        epochs=5, verbose=1)
weights = model.get_weights()
signature = generate_hmac(weights, SECRET_KEY)
local_weights.append((weights, signature))
# Verify client updates
verified_weights = []
for weights, signature in local_weights:
    if verify_hmac(weights, signature, SECRET_KEY):
        verified_weights.append(weights)
    else:
        print("⚠ Tampered client detected!")
if not verified_weights:
    print("✖ No valid clients!")
    break
global_model.set_weights(federated_avg(verified_weights))
loss, acc = global_model.evaluate(X_test, y_test, verbose=0)
history.append(acc)
print("Global Accuracy:", acc)

```

Evaluation and Metrics

```

# Final evaluation
final_loss, final_accuracy = global_model.evaluate(X_test, y_test, verbose=0)
y_pred = (global_model.predict(X_test) > 0.5).astype(int)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
print("\n===== FINAL RESULTS =====")
print("Final Accuracy : {:.2f}%".format(final_accuracy * 100))
print("Precision      : {:.2f}%".format(precision * 100))

```

```
print("Recall      : {:.2f}%".format(recall * 100))
print("F1 Score   : {:.2f}%".format(f1 * 100))
```

Accuracy Graph

```
plt.plot(history, marker='o')
plt.title("Federated Accuracy")
plt.xlabel("Rounds")
plt.ylabel("Accuracy")
plt.grid()
plt.show()
```

Confusion Matrix

```
cm = confusion_matrix(y_test, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm,
    display_labels=["NORMAL", "PNEUMONIA"])
disp.plot(cmap="Blues")
plt.title("Confusion Matrix")
# Save results
output_dir = "results"
os.makedirs(output_dir, exist_ok=True)
plt.savefig(os.path.join(output_dir, "confusion_matrix.png"))
plt.show()
```

CHAPTER 6

RESULTS AND DISCUSSIONS

The performance of the proposed model is analysed by using the performance metrics such as accuracy, precision, recall, F1-score, computation time, latency, and response time. Analysis of accuracy of proposed and existing models is evaluated in Figure 3.

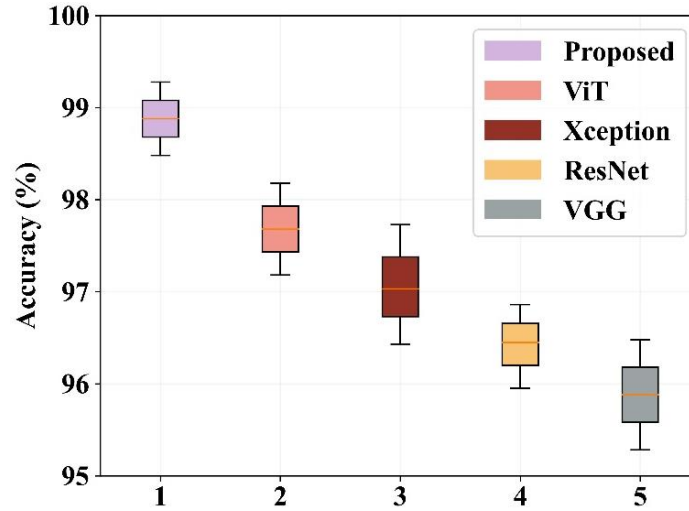


Figure 11: Analysis of accuracy for proposed and existing models

The OSDSTran model attained meaningfully much higher performance compared to other existing models such as VGG [27], ResNet [28], Xception [29], and ViT [30] which reached accuracies of 95.88%, 96.45%, 97.03%, and 97.68% respectively. In contrast, the proposed model obtained an impressive accuracy of 98.88% it will highlighting its effectiveness in demand prediction. This will also reduce the false negatives. Analysis of precision for proposed and existing models is illustrated in Figure 4.

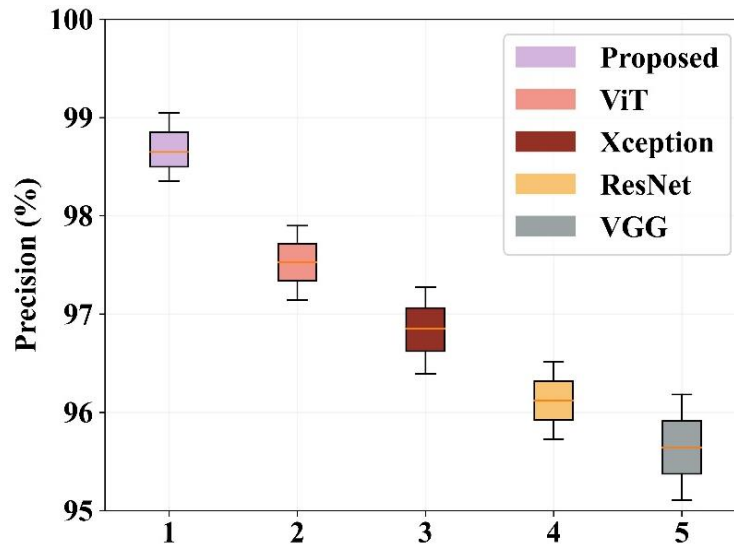


Figure 12: Analysis of precision for proposed and existing models

The figure illustrate the precision of the OSDSTran model attained better than other existing methods. The other models are VGG, ResNet, Xception, and ViT their respective precision rates of 95.64%, 96.12%, 96.85% and 97.53%. Also the proposed OSDSTran model attained a better high precision score of 98.65%. Which significant improvement the proposed model capture more complex features and improve the classification accuracy. Comparison of recall for OSDSTran model and existing models is present in Figure 5.

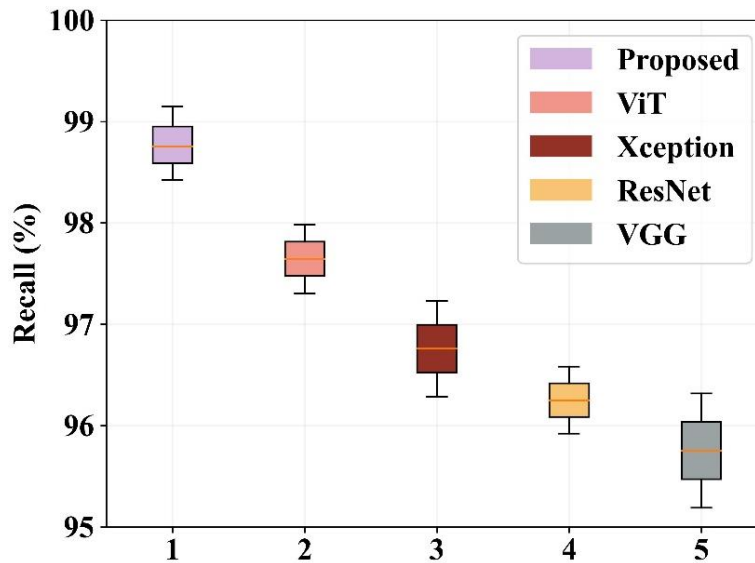


Figure 13: Comparison of recall for OSDSTran model and existing models

The superior performance in recall is shown by the OSDSTran model which scores an amazing 98.75%, considerably outperforming the scores of VGG at 95.75%, ResNet at 96.25%, Xception at 96.75% and ViT at 97.64%. The remarkable higher recall value is accurately identify the models true negatives and false negatives. Analysis of F1-score for proposed model and existing models is show in Figure 6.

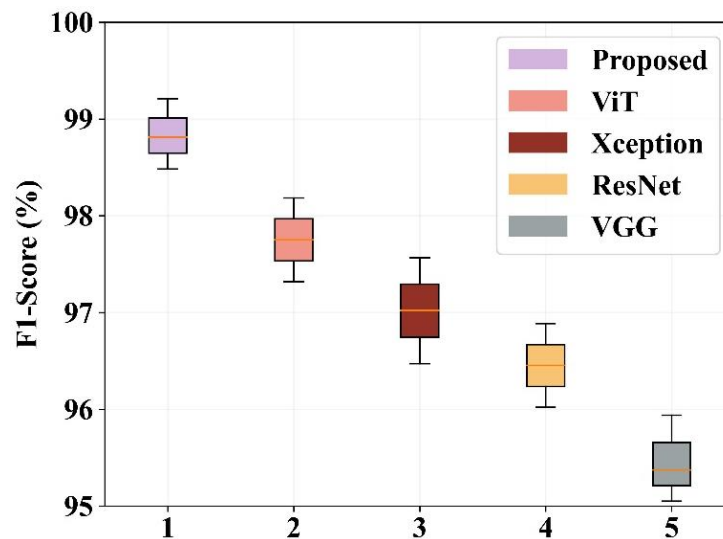


Figure 14: Analysis of F1-score for proposed model and existing models

The graph describes the F1-score for OSDSTran model and several existing models with traditional models such VGG, ResNet, Xception, and ViT scoring 95.37%, 96.45%, 97.02%, and 97.75%, correspondingly. Conversely the suggested OSDSTran model achieved an outstanding F1-score of 98.81%. The high F1-score indicate OSDSTran model well balance in both precision and recall which make system more efficiently. Compression of average computation time for OSDSTran model and existing model is given in Figure 7.

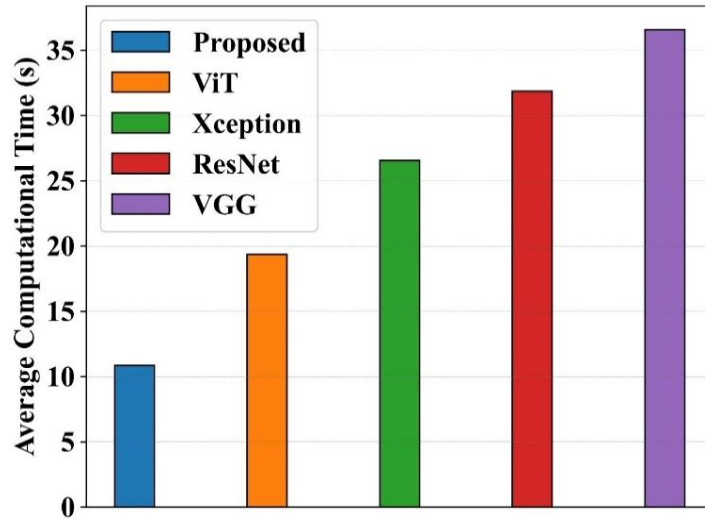


Figure 15: Compression of average computation time for OSDSTran model and existing model

The average computation time of the proposed model is far higher than that of several existing models, such as VGG, ResNet, Xception, and ViT with 36.58 s, 31.85 s, 26.54 s and 19.35 s accuracy values respectively. However the new model ensured an average computation time of 10.85 s. This low average computation time for OSDSTran model which have attained high performance and low computational expensive in real-world processing. Analysis of latency for OSDSTran model and existing model is illustrated in Figure 8.

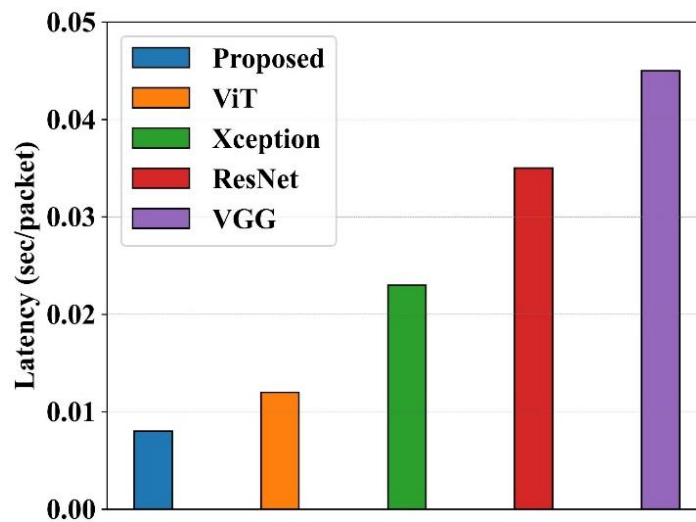


Figure 16: Analysis of latency for OSDSTran model and existing model

The latency of the proposed model considerably exceeds those in many previously constructed models. One such existing models include VGG, ResNet, Xception, and ViT that produced values for 0.045 sec/packet, 0.035 sec/packet, 0.023 sec/packet and 0.012 sec/packet respectively but a new model that emerged perfectly was achieved with a low value of 0.008 sec/packet. Analysis for response time for proposed model is present

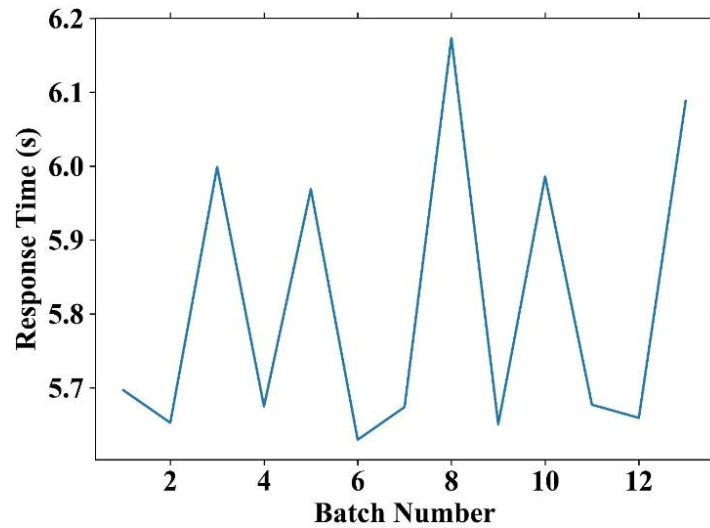


Figure 17: Analysis for response time for proposed model

The response time graph of FL model indicates variation among the various batches, where time varies between 5.5 and 6.1 seconds. Nevertheless, the model has a constant processing time, with periodic spikes, which corresponds to the possible effect of other factors, such as the complexity of the data, delays in communication between local and global systems, or changing computational loads of different batches. Nonetheless, the range of response is stable, and all the variations do not exceed a controllable range, indicating the efficiency of the model in processing IoV data to identify network intrusion. AUC for proposed model is illustrated in Figure 10.

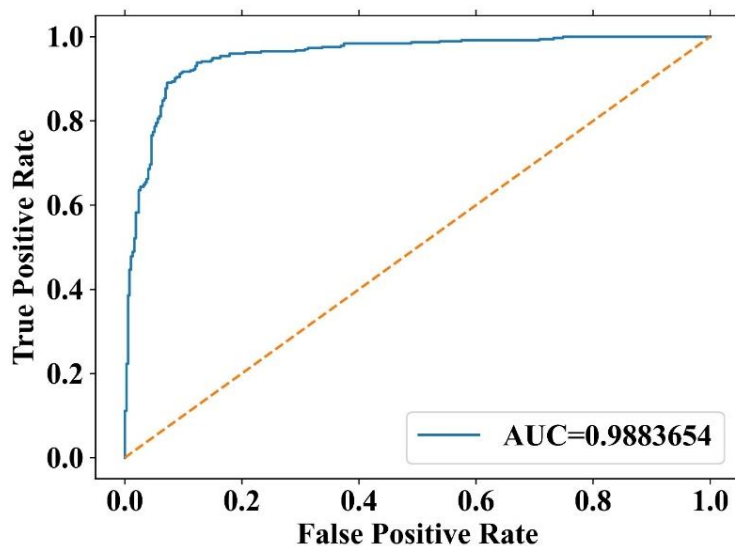


Figure 18: AUC for proposed model

OSDSTran model registers high AUC values of 0.9883 respectively. In addition to demonstrating the models impressive classification accuracy, high AUC highlights its potential applications in industries that require precision in their decisions and risk assessment. It is also a viable real-time processing tool because it enhances the performance of the system. PRC for OSDSTran model is illustrated in Figure 11.

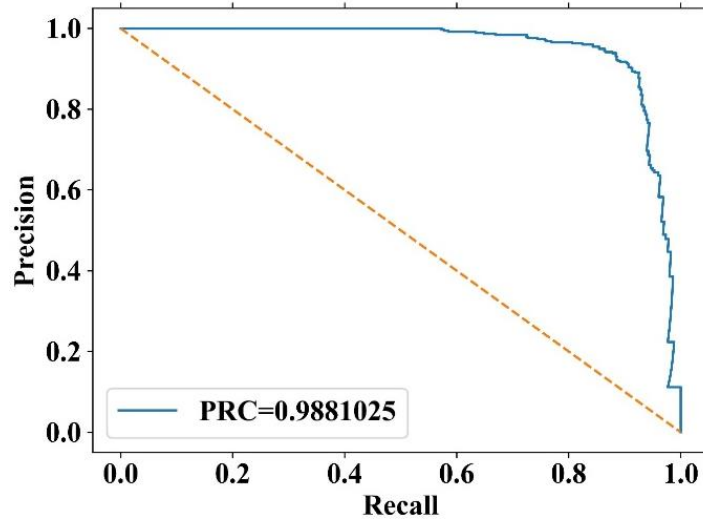
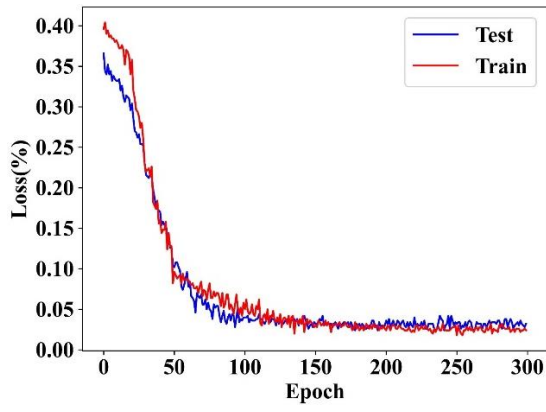
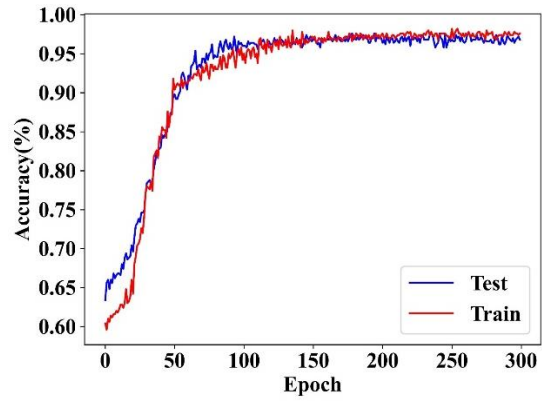


Figure 19: PRC for OSDSTran model

The OSDSTran model is recorded high PRC values of 0.9881 respectively. Beyond showing the models remarkable classification accuracy, this high PRC emphasizes its possible for practical uses in sectors needing exact decision-making and risk analysis. It is also a useful tool for real-time processing as it improves system performance. Analysis of training, testing loss and accuracy of the proposed model is represented in the Figure 12.



(a)



(b)

Figure 20: Analysis of training, testing loss and accuracy of the proposed model

CHAPTER 7

CONCLUSION

7.1 Conclusion

The proposed privacy-preserving healthcare device authentication system using federated learning successfully addresses major challenges related to data security and privacy in IoMT environments. By integrating a dual-layer authentication mechanism using HMAC-SHA256 and the Centralized Authorization Authority (CAA), the system ensures that only authorized devices participate in the learning process. The use of the OSDSTran model significantly improves performance, achieving high accuracy and precision compared to existing models. Additionally, federated learning enables local data processing, ensuring that sensitive medical data is not shared, thereby maintaining patient privacy. Overall, the system provides a secure, efficient, and reliable solution for healthcare data analysis.

7.2 Future Scope

In the future, the proposed system can be enhanced by incorporating more advanced security techniques such as blockchain for improved trust and transparency. The model can also be extended to handle larger and more diverse medical datasets for better generalization in real-time applications. Further improvements can be made by reducing computational complexity and optimizing communication efficiency between devices. Additionally, the system can be integrated with real-time healthcare monitoring systems and wearable devices for continuous patient analysis. Expanding the framework to support multi-disease detection and other healthcare domains can further increase its practical applicability.

REFERENCES:

1. Khan, Habib Ullah, Yasir Ali, and Faheem Khan. "A features-based privacy preserving assessment model for authentication of internet of medical things (IoMT) devices in healthcare." *Mathematics* 11, no. 5 (2023): 1197.
2. Das, Sangjukta, and Suyel Namasudra. "Lightweight and efficient privacy-preserving mutual authentication scheme to secure internet of things-based smart healthcare." *Transactions on Emerging Telecommunications Technologies* 34, no. 11 (2023): e4716.
3. Ryu, Hyunho, and Hyunsung Kim. "Privacy-preserving authentication protocol for wireless body area networks in healthcare applications." In *Healthcare*, vol. 9, no. 9, p. 1114. MDPI, 2021.
4. Azad, Muhammad Ajmal, Junaid Arshad, Shazia Mahmoud, Khaled Salah, and Muhammad Imran. "A privacy-preserving framework for smart context-aware healthcare applications." *Transactions on Emerging Telecommunications Technologies* 33, no. 8 (2022): e3634.
5. Soni, Mukesh, and Dileep Kumar Singh. "Privacy-preserving authentication and key-management protocol for health information systems." In *Data Protection and Privacy in Healthcare*, pp. 37-50. CRC Press, 2021.
6. Rajasekaran, Arun Sekar, Azees Maria, Maheswar Rajagopal, and Josip Lorincz. "Blockchain enabled anonymous privacy-preserving authentication scheme for internet of health things." *Sensors* 23, no. 1 (2022): 240.
7. Almaiah, Mohammed Amin, Fahima Hajjej, Aitizaz Ali, Muhammad Fermi Pasha, and Omar Almomani. "A novel hybrid trustworthy decentralized authentication and data preservation model for digital healthcare IoT based CPS." *Sensors* 22, no. 4 (2022): 1448.
8. Kathamuthu, Nirmala Devi, Annadurai Chinnamuthu, Nelson Iruthayanathan, Manikandan Ramachandran, and Amir H. Gandomi. "Deep Q-learning-based neural network with privacy preservation method for secure data transmission in internet of things (IoT) healthcare application." *Electronics* 11, no. 1 (2022): 157.
9. Kute, Shruti, A. V. Shreyas Madhav, Amit Kumar Tyagi, and Atharva Deshmukh. "Authentication Framework for Healthcare Devices Through Internet of Things and Machine Learning." In *Evolutionary Computing and Mobile Sustainable Networks: Proceedings of ICECMSN 2021*, pp. 399-415. Singapore: Springer Singapore, 2022.
10. Singh, Saurabh, Shailendra Rathore, Osama Alfarraj, Amr Tolba, and Byungun Yoon. "A framework for privacy-preservation of IoT healthcare data using Federated Learning and blockchain technology." *Future Generation Computer Systems* 129 (2022): 380-388.
11. Ali, Mansoor, Faisal Naeem, Muhammad Tariq, and Georges Kaddoum. "Federated learning for privacy preservation in smart healthcare systems: A comprehensive survey." *IEEE journal of biomedical and health informatics* 27, no. 2 (2022): 778-789.
12. Sabry, Farida, Tamer Eltaras, Wadha Labda, Khawla Alzoubi, and Qutaibah Malluhi. "Machine learning for healthcare wearable devices: the big picture." *Journal of Healthcare Engineering* 2022, no. 1 (2022): 4653923.
13. Li, Xiong, Jiabei He, Pandi Vijayakumar, Xiaosong Zhang, and Victor Chang. "A verifiable privacy-preserving machine learning prediction scheme for edge-enhanced HCPSs." *IEEE Transactions on Industrial Informatics* 18, no. 8 (2021): 5494-5503.
14. Soni, Mukesh, and Dileep Kumar Singh. "Privacy-preserving authentication and key-management protocol for health information systems." In *Data Protection and Privacy in Healthcare*, pp. 37-50. CRC Press, 2021.
15. Unal, Devrim, Shada Bennbaia, and Ferhat Ozgur Catak. "Machine learning for the security of healthcare systems based on Internet of Things and edge computing." In *Cybersecurity and cognitive science*, pp. 299-320. Academic Press, 2022.

16. Ahmed, Syed Thouheed, A. C. Kaladevi, Achyut Shankar, and Fayez Alqahtani. "Privacy enhanced edge-ai healthcare devices authentication: a federated learning approach." *IEEE Transactions on Consumer Electronics* (2025).
17. Ayyappan, G., V. Loganathan, E. Padma, S. Ilavarasan, and A. Subash. "Federated Learning and Edge AI for Privacy-Preserving Diabetes Prediction in Healthcare." In *2025 3rd International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS)*, pp. 1116-1121. IEEE, 2025.
18. Yang, Xinyi, and Juan Li. "A clustering-based federated deep learning approach for enhancing diabetes management with privacy-preserving edge artificial intelligence." *Healthcare Analytics* 7 (2025): 100392.
19. Eshwarappa, Nithin Melala, Hojjat Baghban, Ching-Hsien Hsu, Po-Yen Hsu, Ren-Hung Hwang, and Mu-Yen Chen. "Communication-efficient and privacy-preserving federated learning for medical image classification in multi-institutional edge computing." *Journal of Cloud Computing* 14, no. 1 (2025): 44.
20. Chen, Ke, and Long Xiao. "Energy-efficient privacy-preserving AI models for real-time health monitoring in mobile IoT networks for professional sports applications." *Discover Applied Sciences* 7, no. 9 (2025): 1-25.
21. Choi, Hojin, and Seog Chung Seo. "Optimization of PBKDF2 using HMAC-SHA2 and HMAC-LSH families in CPU environment." *IEEE Access* 9 (2021): 40165-40177.
22. Lin, Ailiang, Bingzhi Chen, Jiayu Xu, Zheng Zhang, Guangming Lu, and David Zhang. "Ds-transunet: Dual swin transformer u-net for medical image segmentation." *IEEE Transactions on Instrumentation and Measurement* 71 (2022): 1-15.
23. Zhang, Xiangrong, Shouwang Shang, Xu Tang, Jie Feng, and Licheng Jiao. "Spectral partitioning residual network with spatial attention mechanism for hyperspectral image classification." *IEEE transactions on geoscience and remote sensing* 60 (2021): 1-14.
24. Biswas, Saptadeep, Azharuddin Shaikh, Absalom El-Shamir Ezugwu, Japie Greeff, Seyedali Mirjalili, Uttam Kumar Bera, and Laith Abualigah. "Enhanced prairie dog optimization with Levy flight and dynamic opposition-based learning for global optimization and engineering design problems." *Neural Computing and Applications* 36, no. 19 (2024): 11137-11170.
25. Zhou, Yuhao, Qing Ye, and Jiancheng Lv. "Communication-efficient federated learning with compensated overlap-fedavg." *IEEE Transactions on Parallel and Distributed Systems* 33, no. 1 (2021): 192-205.
26. <https://www.kaggle.com/datasets/nih-chest-xrays/data>
27. Kareem, Amer, Haiming Liu, and Vladan Velisavljevic. "A privacy-preserving approach to effectively utilize distributed data for malaria image detection." *Bioengineering* 11, no. 4 (2024): 340.
28. Adnan, Amna, Firdous Kausar, Muhammad Shoaib, Faiza Iqbal, Ayesha Altaf, and Hafiz M. Asif. "A Secure and Privacy-Preserving Approach to Healthcare Data Collaboration." *Symmetry* 17, no. 7 (2025): 1139.
29. Patel, Nisarg P., Raj Parekh, Saad Ali Amin, Rajesh Gupta, Sudeep Tanwar, Neeraj Kumar, Rahat Iqbal, and Ravi Sharma. "LEAF: A federated learning-aware privacy-preserving framework for healthcare ecosystem." *IEEE Transactions on Network and Service Management* 21, no. 1 (2023): 1129-1141.
30. Amin, Al, Kamrul Hasan, Sharif Ullah, and M. Shamim Hossain. "ViT Enhanced Privacy-Preserving Secure Medical Data Sharing and Classification." In *2025 IEEE 22nd Consumer Communications & Networking Conference (CCNC)*, pp. 1-2. IEEE, 2025.

PAPER PUBLICATION

Privacy-Preserving Healthcare Device Authentication via Edge-AI and Federated Learning

Abstract

Internet of Medical Things (IoMT) and hospital-edge devices to monitor patient health have accelerated the process of generating sensitive medical data. Although this type of data can make a significant contribution to the detection of the disease with the help of artificial intelligence (AI) but also traditional model faced several limitation such as data privacy, computational cost and over-fitting issues. To overcome the limitation this paper suggests a privacy-preserving Federated Learning (FL) model that incorporates a token-based dual-layer authentication and high-level medical image classification. The suggested system requires that hospitals will initially be registered in a Centralized Authorization Authority (CAA) where metadata of devices and hospital credentials are checked and HMAC-SHA256 tokens are generated. FL only accepts authenticated hospitals that possess active tokens, and rest of the nodes are not allowed, therefore safe and reliable participation is guaranteed. To train locally, each hospital applies Optimized Spatial Densely Connected Swin Transformer (OSDSTran), an architecture with a combination of spatial attention and transformer layers to successfully discern local and global characteristics of the chest X-ray images to classify the disease. The Prairie Dog Optimization Algorithm (PDOA) is then used to refine the hyper-parameters and to provide an accurate result. Authorized updates are fully secured by FedAvg, and the patient information is all local, which guarantees a high level privacy protection. The proposed method has been proven to be efficient in secure and reliable medical image classification with accuracy of 98.88% and precision of 98.65%, which will improve the model performance in real-time.

Keywords: Healthcare Device, Internet of Medical Things, Authentication, Federated Learning, Swin Transformer, Centralized Authorization Authority, Prairie Dog Optimization, FedAvg

1. Introduction

Internet of Things (IoT) devices have gained considerable attention in the recent past years, as they have had a substantial contribution to all aspects of life, yet the greatest implementation of the devices has been recorded in the healthcare sector [1]. The modern healthcare system has become automated, efficient, and smart due to the use of IoT and other technologies such as cloud computing and machine learning [2]. It assists medical workers to receive healthcare data of patient remotely and offers remote medical diagnosis. The data being more of a privacy concern in the health sector must offer privacy-sensitive, and in this context, security and privacy must be taken into account [3]. On the one hand, these devices help to enhance the quality of healthcare, but on the other hand, they produce data that includes sensitive information about individual patients [4]. Where the unauthorized access will amount to the violation of privacy with disastrous consequences to an individual and financial damages to the ruling authority through a regulation like the General Data Protection Regulation [5].

The IoHT technology is not costly and involves the non-invasive medical devices. In addition, an IoHT is a significant part of mobile health monitoring systems and radically enhances the healthcare quality and efficiency [6]. The development of the blockchain and artificial intelligence technology also facilitates the development of safe intelligent healthcare [7]. The intelligence ML technologies have been used to develop the IoHT to enhance and support patient care and healthcare services through the abundance of data produced by the connected medical equipment [8]. As a rule, ML is a field that encompasses both the two significant disciplines as building intelligent computer systems and controlling these systems [9]. In federated learning, the ML algorithms are trained on the local devices/machines or mobile phones without the need of transferring the data on the local devices to the central point such

as clouds [10].

Nevertheless, the centralized training philosophical consideration of artificial intelligence (AI) causes privacy concerns related to the information exchanged between mobile and wearable IoMT devices and hospitals as well as end-users [11]. It may be highly applicable in several healthcare and geriatric care applications like activities detection to determine the health state, fall detection, stress detection, fitness tracking, vital signs monitoring, and the diagnosis of diseases [12]. The use of the ML prediction towards users of human cyber-physical systems (HCPSs) offers predictable results of prediction, using matured models, but the security and privacy protection are significant concerns [13]. IoMT has faced various security issues pertaining to the tracking of the healthcare data of patients [14]. The rising rate of cyber-attacks of hospitals systems, and the fact that mathematical operations on data stored on clouds are being exposed to untrusted parties is a great threat to the data privacy and safety [15]. To address these limitation proposed model present a new deep learning model.

1.1 Motivation and problem statement

Medical devices connected via Internet are open to cyber-attacks, hence the privacy preservation of the user and the devices plays an important role. The current model has a number of remarkable limitations. The time spent in the authentication process can be decreased through frequent authentication, and heterogeneous hospital nodes or IoT devices typically require massive local training, which can lead to more computation time and its expenses. High-quality images are utilized in the model which impacts its dependability in real-time clinical use. Also, the model has a drawback in that it cannot be generalized due to its overall complexity, which may result in poor performance in a variety of situations. The current paper presents the proposal of Medical Edge-devices privacy enhanced technique on the basis of the federated learning principles on the distributed coordination of devices and computational system. The trend of using IoMT and hospital-edge devices to monitor patients creates sensitive medical information that cannot be centrally shared because of privacy policies, and thus, using traditional AI training is risky. Standard FL guarantees locality of data but does not provide the strength of authentication and the global model is susceptible to updates by malicious or unauthorized users. This paper is driven by the desire to have secure, privacy-sensitive, and trustworthy AI in the healthcare field and consequently, it combines token-based, dual-layer authentication and federated learning to make sure that only the verified hospitals feed into the global model and the model is highly accurate and the patient information remains confidential. The major contribution of this work is given below.

- To implement a HMAC-SHA256 based authentication mechanism through the Centralized Authorization Authority (CAA) for verifying hospitals.
- To develop an Optimized Spatial Densely Connected Swin Transformer (OSDSTran) for classify the diseases and capture the important features.
- To suggest a Prairie Dog Optimization Algorithm (PDOA) for enhancing the model classification performance

The paper is organized from Section 1, which consists of the introduction. Section 2 consists of related works, Section 3 consists of the proposed methodology, and Section 4 consists of the result and discussion of the proposed model. Finally, the conclusion and future work for the proposed model is provided in Section 5.

2. Related Work

Some of the recent works on Privacy-Preserving Healthcare Device Authentication federated learning and deep learning approaches are described as follows.

Ahmed et al. [16] developed Medical Edge-devices privacy enhanced technique based on the federated learning for devices authentication. CAA creates special tokens to active participants of the federated learning process. In order to ensure improved security, a two-tier token validation scheme is implemented, comprising the Authentication Authority Valuation Layer (AAVL) and the Centralized Server Firewall Authentication Layer (CSFAL) which undertake the issuance, validation and revocation of tokens respectively. The model achieved highest accuracy of 96%. The model may suffer reduced operational time due to the frequent authentication.

Ayyappan et al. [17] constructed federated edge AI for privacy-preserving diabetes prediction. The framework proposed incorporates federated learning such that the healthcare institutions and IoT-enabling edge devices do not share raw patient data, and instead, a central server uses Federated Averaging (FedAvg) to incorporate encrypted local updates into a global model. In the experimental evolution model attained the accuracy of 91.2%. Different hospital nodes or IoT devices may requires high training.

Yang et al. [18] implemented clustering-based federated deep learning (Clu-FDL) for privacy-preserving edge AI. The suggested model Clu-FDL, is a hybrid of federated learning and a personalized clustering plan that ensures the grouping of patients based on their carbohydrate (CHO) intake patterns and allows custom model training without involved in sharing of sensitive data. The framework uses simple recurrent neural networks (SimpleRNN) in order to model temporal dependencies in glucose variations. The model attained the highest precision of 93%. Clu-FDL model required high computational power in training time this may increase computational cost.

Eshwarappa et al. [19] presented Edge-Average Federated Learning (Edge-Avg) for skin cancer image classification. The suggested Edge-Avg approach incorporates edge intelligence training in a multi-institutional setting, in which local healthcare facilities are training deep learning to classify skin cancer images without exchanging raw data, hence saving patient privacy. In a bid to solve the communication and computational latency, the framework proposes a better Edge-Avg aggregation algorithm, one that is more efficient in training than the conventional Fed-Avg. Fed-Avg method by achieving a better accuracy of 93.94%. The dependence on image quality was affected in reliability of real-time clinical use.

Chen et al. [20] suggested Adaptive Quantized Federated Dual-Network Framework (AQ-FDNF) for real-time health monitoring in mobile IoT networks. The AQ-FDNF framework proposed consists of a hybrid deep neural network that incorporates Temporal Convolutional Networks (TCN) and attention-enhanced Gated Recurrent Units (A-GRU) to extract temporal bio-signals the features and decode dynamic sequences respectively, both with the help of layer-wise adaptive quantization to make inference in heterogeneous mobile IoT wearable's energy-efficient. The model attained the accuracy of 93.8%. The complexity of the model was limited generalization issues this may degrade model performance. The analysis of existing models with its performance and disadvantages is presented in Table 1.

Table 1: The analysis of existing models with its performance and disadvantages

Author name & References	Models	Performance	Disadvantages
Ahmed et al. [16]	Medical Edge-devices privacy enhanced technique based on the federated learning	Accuracy of 96%	The model may suffer reduced operational time due to the frequent authentication.
Ayyappan et al. [17]	Federated edge AI	Accuracy of 91.2%	Different hospital nodes or IoT devices may requires high training.
Yang et al. [18]	Clu-FDL	Precision of 93%	Clu-FDL model required high computational power in training time this may increase computational cost.
Eshwarappa et al.	Edge-Avg	Accuracy of 93.94%	The dependence on

[19]			image quality was affected in reliability of real-time clinical use.
Chen et al. [20]	AQ-FDNF	Accuracy of 93.8%	The complexity of the model was limited generalization issues this may degrade model performance.

The existing model faced several disadvantages such as may suffer reduced operational time due to the frequent authentication. Additionally different hospital nodes or IoT devices may require high training. Moreover model required high computational power in training time this may increase computational cost. Furthermore dependence on image quality was affected in reliability of real-time clinical use. Finally the complexity of the model was limited generalization issues this may degrade model performance.

3. Proposed Methodology

Medical devices connected via Internet are open to cyber-attacks, hence the privacy preservation of the user and the devices plays an important role. In this paper, Medical Edge-devices privacy enhanced technique based on the federated learning principles is proposed on the distributed coordination of devices and computational system. Workflow diagram of proposed model is illustrated in Figure 1.

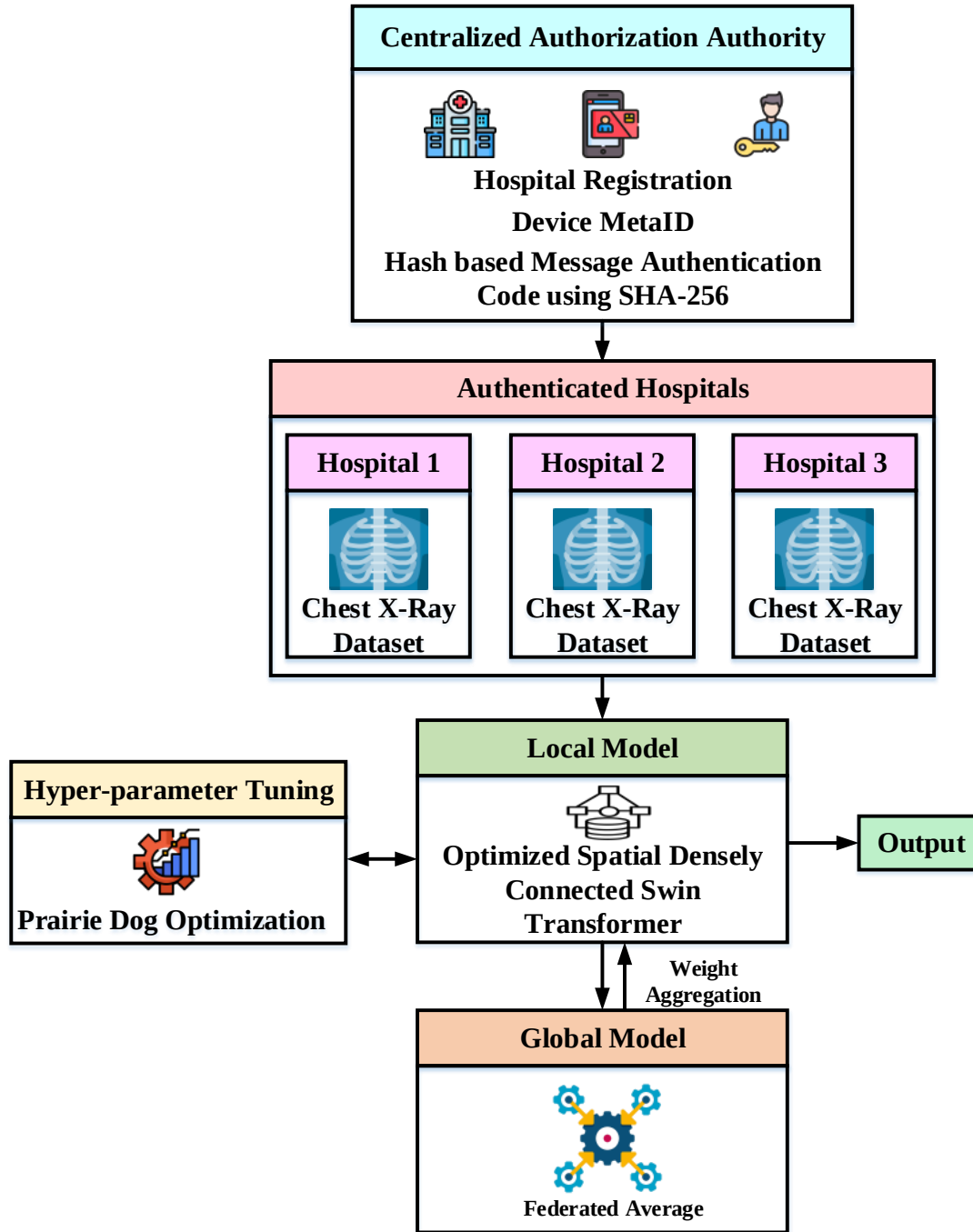


Figure 1: Workflow diagram of proposed model

In the proposed privacy-preserving **Federated Learning framework**, each hospital first registers with the centralized authorization authority (CAA), where devices and hospitals credentials (Hospital ID, device metadata) are recorded and **Hash-based Message Authentication Code using SHA-256 (HMAC-SHA256)** is issued for authentication. Active tokens are required, and hospitals are the part of FL network are allowed to exclude the unauthorized or compromised nodes. After authentication, individual hospitals load its local medical dataset, that is, Chest X ray and local model training through “**Optimized spatial densely connected Swin transformer based network (OSDSTran)**” to classify the diseases. Spatial attention in medical image classification is used to selectively focus on the most informative regions and transformer layer is efficiently captures the local and global features, making it highly effective for accurate diseases classification. The hyper-parameters are tuned by using “**Prairie dog optimization Algorithm (PDOA)**” which will improve the classification accuracy. This method store the raw data locally to maintain patient privacy. The authentication mechanism verifies the hospital according to the data stored in **CAA**. Only validated updates

are submitted to the central server to be securely aggregated, by **FedAvg**. Lastly, the authenticated local model updates are classified to improve privacy.

3.1 Hash-based Message Authentication Code using SHA-256

The centralized authorization authority (CAA), where devices and hospitals credentials (Hospital ID, device metadata) are recorded and HMAC-SHA256 is issued for authentication. An HMAC can also be computed using any cryptographic hash function, e.g. SHA-2 or SHA-3, and the combined algorithm is usually represented as HMAC-X, where X is the hash function used. HMAC security HMAC is cryptographically secure subject to a number of factors: the security properties of the underlying hash function, the size of the hash output and the quality and length of the secret key. The HMAC works in two hash passes. The secret key is firstly processed to obtain two different keys which are the inner and outer key. The inner key is used in the first pass, in conjunction with the message to get an intermediate hash value. The outer key and this middle hash are then inputted into the second pass which then produces the final HMAC code [21]. The two-step process increases the level of security especially in offering resistance to attacks on length extension. The underlying hash function is iterative, enabling it to handle the input message in fixed sized blocks. The length of the output of the HMAC hash is the same as the underlying hash functionality, yet it may be truncated. In transmitting the data, the message is delivered along with a HMAC tag. The other party who is receiving the message and has access to the secret key recomputed the HMAC of the message received to confirm its integrity and authenticity.

$$HMAC(K, b) = L((K' \oplus Q_{outer}) \| L((K' \oplus Q_{inner}) \| b)) \quad (1)$$

$$K' = \begin{cases} L(K), & \text{For } K \text{ value higher than the block size} \\ K, & \text{otherwise} \end{cases} \quad (2)$$

Equation (1) represents the authenticated message which is represented by b , the hash function which is represented by L , secret key which is represented by K , and block-sized key which is represented by K' . The inner and outer padding of block size are denoted as Q_{inner} and Q_{outer} respectively.

3.2 Optimized spatial densely connected Swin transformer based network

After authentication, individual hospitals load its local medical dataset, that is, Chest X ray and local model training through Optimized spatial densely connected Swin transformer based network (OSDSTran) to classify the diseases. The standard transformer encoder is made up of a stack of L identical blocks. Every block consists of a Multi-head Self-Attention (MSA) module and a Multi-Layer Perceptron (MLP) [22]. Also there is a Layer Normalization (LN) layer before the MSA and MLP modules and a residual connection placed after each of the modules. Therefore the Transformer encoder in position can be represented as follows.

$$\hat{k}_m = MSA(LN(k_{m-1})) + k_{m-1} \quad (3)$$

$$k_m = MLP(LN(\hat{k}_m)) + \hat{k}_m \quad (4)$$

Under the conventional transformer architecture, the correlations of each token with all other tokens are to be calculated, which has a quadratic relationship with the number of tokens. This renders the method not very practical to most dense prediction tasks and processing of high-resolution images. Architecture of 2 Optimized spatial densely connected Swin transformer based network for proposed model is show in Figure 2.

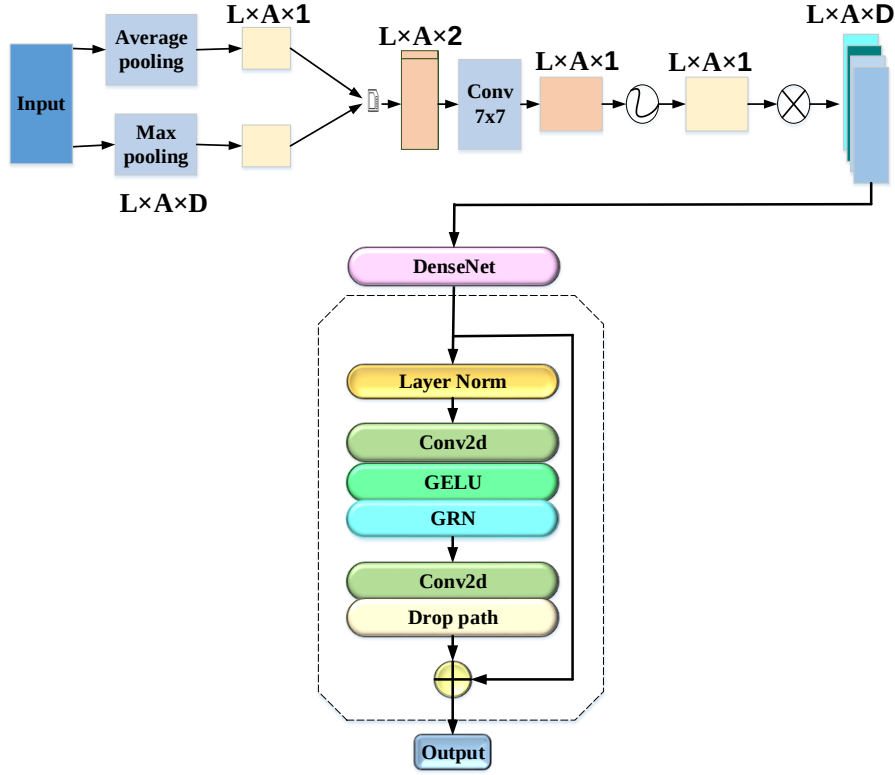


Figure 2: Optimized spatial densely connected Swin transformer based network for proposed model

To solve this Swin transformer proposes Window-based Multi-head Self-Attention (W-MSA) and Shifted Window-based Multi-head Self-Attention (SW-MSA) mechanisms [23]. In W-MSA all the patches of the input feature map are separated into non-overlapping windows which have $N \times N$ patches. Only within these local windows self-attention is then carried out greatly lowering the cost of computation. The output of the W-MSA, and the MLP in the m th layer are represented as $\hat{k}_m$ and k_m determined as follows.

$$\hat{k}_m = W - MSA(LN(k_{m-1})) + k_{m-1} \quad (5)$$

$$k_m = MLP(LN(\hat{k}_m)) + \hat{k}_m \quad (6)$$

One of the weaknesses of W-MSA is that the information exchange among various windows cannot be realized effectively. To overcome this, without raising computation expenses SW-MSA is proposed and implemented after W-MSA. SW-MSA window structure is also different as compared to that of the previous W-MSA layer where it uses an effective batch processing method that uses cyclic shifting of the feature map towards the upper-left. This change means that each batch window is a collection of several sub-windows in the feature map that are not adjacent to each other, but all in the same quantity as in the standard partitioning. Notwithstanding the fact that the calculation of self-attention remains local in each window in both W-MSA and SW-MSA, there is an addition of a relative position bias when computing similarity scores. In this rearranged window partitioning method, the results of the SW-MSA and the successive MLP module can be formulated as shown below.

$$\hat{k}_{m-1} = SW - MSA(LN(k_m)) + k_m \quad (7)$$

$$k_{m+1} = MLP(LN(\hat{k}_{m-1})) + \hat{k}_{m-1} \quad (8)$$

Spatial Attention Module (SAM) was proposed in the convolutional block attention module, which is used in classification and detection problems. SAM uses the spatial relation between features to come up with a spatial attention map. In computing this spatial attention, SAM first performs max-pooling operation and average-pooling operation along the channel axis after which the feature maps obtained are concatenated with each other to create an efficient feature descriptor. Informally, an input feature denoted by G , is inputted into channel-wise max-pooling and average-pooling layers to create outputs, denoted by $G_{mp}^T \in S^{O \times A \times 1}$ and $G_{ap}^T \in S^{O \times A \times 1}$.

, respectively. These two concatenated feature maps are then inputted into a convolutional layer and then a sigmoid activation to produce the spatial attention map $NM^T(G) \in S^{O \times A \times 1}$. Overall, the output of the spatial attention module F_s is represented by a set of numbers $G^T \in S^{O \times A \times D}$ of the form.

$$\begin{aligned} G^T &= G.NM^T(G) \\ &= G.\sigma(g^{7 \times 7}([Maxpool(G); Avgpool(G)])) \\ &= G.\sigma(g^{7 \times 7}([G_{mp}^T; G_{ap}^T])) \end{aligned} \quad (9)$$

Here $g^{7 \times 7}(\cdot)$ represents the convolution operation with a kernel size of 7 and $\sigma(\cdot)$ denotes the sigmoid function.

Proposed a solution to vanishing gradient problem during deep convolutional networks training with the ResNet (Residual Network). ResNet makes it easy to train significantly deeper networks than those utilized before by adding skip connections to avoid nonlinear transformations and instead to have identity mappings. One of the benefits of ResNet is that one can pass the gradients through these identity connections between previous layers and the subsequent ones, enhancing training stability. Furthermore the facilitate information flow among layers is a connectivity pattern known as DenseNet (Densely Connected Convolutional Networks). Instead of extreme deep or wide architecture to add representational power, DenseNet uses feature reuse to create smaller models that are simpler to train and are characterised by high parameter efficacies. DenseNet enhances the variation of inputs to the next layer, and in general, the efficiency of the network by linking every layer to all other layers. This high connectivity is a significant difference between DenseNet and ResNet. Take one input feature is i_0 through a convolutional network with ml layers. A layer x has a nonlinear transform $J_{ml}(\cdot)$. Therefore the feature maps of all the earlier layers are passed on as input to the ml layer $i_0, i_1 \dots i_{ml}$.

$$i_{ml} = J_{ml}[(i_{ml-1}, i_{ml-2}, i_{ml-3}, \dots, i_0)] \quad (10)$$

As a means to construct a powerful network, visual attention has been utilized. Connection-wise attention model proposed in this paper allows the network to devote visual attention to the feature maps of each layer. The proposed OSDSTran is combines with the feature maps of all the previous layers, the DenseNet which concatenates the features of all the previous layers with the next layer. The weighting parameters are trained during the network training so the network can focus on the most contributory information. This method eases and enhances efficiency of the network. The OSDSTran uses a weighting coefficient AW_x in the x th layer of the network. The equation (12) shows the layout of the connection-wise-attention where $J_{ml}(\cdot)$ a non-linear transformation is, and where $i_y (1 \leq y \leq ml - 1)$ is the feature-maps of the y th layer.

$$AW_x = [aw_{x-1,x}, aw_{x-2,x}, \dots, aw_{2,x}, aw_{1,x}] \quad (11)$$

$$i_{ml} = J_{ml}(aw_{ml-1,ml}i_{ml-1}) + aw_{ml-2,ml}i_{ml-2} \dots + aw_{1,ml}i_1 \quad (12)$$

Four types of layers were present in the network. The input layer was the first type, where patches of images were entered into the network. The second one was the convolutional layer, which operated on the input features by applying learned filters to each image to create feature maps of each filter. The OSDSTran was built with 9 convolutional layers with each pair of convolutional layers being followed by max pooling layer. The size of the convolutional filters was $3 \times 3 \times 3$. The feature maps were max pooled to decrease the spatial information of the feature maps by substituting each area with its maximum value maintaining the most effective ones to discriminate features. Finally the model OSDSTran classify the result.

3.3 Prairie dog optimization Algorithm

A colony of prairie dogs (PD) with a number of PD is o and coteries is denoted by n . As PD live and act as a group in a coterie, a position of the individual prairie dog in a particular coterie can be described as a vector [24]. The locations of all coteries (AC) in the colony are depicted in the matrix illustrating the colony represented in equation (13).

$$AC = \begin{bmatrix} AC_{1,1} & AC_{1,2} & \cdots & AC_{1,e-1} & AC_{1,e} \\ AC_{2,1} & AC_{2,2} & \cdots & AC_{2,e-1} & AC_{2,e} \\ \vdots & \vdots & & \vdots & \vdots \\ AC_{n,1} & AC_{n,2} & \cdots & AC_{n,e-1} & AC_{n,e} \end{bmatrix} \quad (13)$$

In this case $AC_{x,y}$ denotes the y th dimension of coterie x th in the colony. Equation (14) depicts the positions of all prairie dogs in a particular coterie.

$$PD = \begin{bmatrix} PD_{1,1} & PD_{1,2} & \cdots & PD_{1,e-1} & PD_{1,e} \\ PD_{2,1} & PD_{2,2} & \cdots & PD_{2,e-1} & PD_{2,e} \\ \vdots & \vdots & & \vdots & \vdots \\ PD_{o,1} & PD_{o,2} & \cdots & PD_{o,e-1} & PD_{o,e} \end{bmatrix} \quad (14)$$

Where $PD_{x,y}$ denotes the y th dimension of the x th prairie dog in a coterie with an assumption that o is much greater than n . The sites of both coterie and prairie dogs are set on a uniform distribution as shown in equation (15) and (16) respectively.

$$AC_{x,y} = V(0,1) \times (UPB_y - LWB_y) + LWB_y \quad (15)$$

$$PD_{x,y} = V(0,1) \times (upb_y - lwb_y) + lwb_y \quad (16)$$

Let UPB_y and LWB_y be the upper and lower limits of the y th dimension of the optimization problem respectively. Let them be defined by $upb_y = \frac{UPB_y}{n}$, with a constant n . Additionally

will use the notation $V(0,1)$ to refer to a uniformly distributed random number in the range 0 and 1. Fitness value of the position of each PD is calculated by entering the required solution vector, in the given fitness function. The resulting values of fitness are then stored in the array as follows in equation (17).

$$g(PD) = \begin{bmatrix} g_1([PD_{1,1} \ PD_{1,2} \ \cdots \ PD_{1,e-1} \ PD_{1,e}]) \\ g_2([PD_{2,1} \ PD_{2,2} \ \cdots \ PD_{2,e-1} \ PD_{2,e}]) \\ \vdots \\ g_o([PD_{o,1} \ PD_{o,2} \ \cdots \ PD_{o,e-1} \ PD_{o,e}]) \end{bmatrix} \quad (17)$$

Fitness values of individual PD reflects the quality of food in a particular site, to build new burrows and the effectiveness of responding to anti-predator cues. These fitness values are arranged in an array and the lowest fitness value is chosen as the optimal solution available till now to the minimization problem at hand. Also, the three lowest values of fitness are taken in combination with the optimality value to inform burrow development that assists prairie mice to evade predators. This section presents the exploratory mechanism of the PD optimization algorithm. The foraging and construction of burrows behaviours of the prairie dogs are used to search the space of optimization problem. When the food sources have been found, prairie dogs inform the other members of their findings by using special sounds. The quality of any given food source is evaluated and the best source is taken up for foraging and new burrows are built depending on the quality of the food source. Equation (18) provides the update of the position of foraging during the exploration phase of the algorithm. The second plan is the quality assessment of the found sources of food and the power of digging. Based on this digging strength, new burrows are constructed, and they are supposed to reduce with the increase in the number of iterations. This is a mechanism that is used to restrict the burrows created with time. Equation (19) provides the position update of burrow building.

$$PD_{x+1,y+1} = Hbest_{x,y} - eDbest_{x,y} \times \rho - ACE_{x,y} \times Levy(o) \forall iter < \frac{Max_{iter}}{4} \quad (18)$$

$$PD_{x+1,y+1} = Hbest_{x,y} \times SPD \times CDS \times levy(o) \forall \frac{Max_{iter}}{4} \leq iter < \frac{Max_{iter}}{2} \quad (19)$$

Where $Hbest_{x,y}$ signifies the global best solution to date. The phrase $eDbest_{x,y}$ indicates the

impact of the best solution now as it is defined in the equation (20). The parameter ρ is the specialized food source alarm frequency which was constant at 0.1 kHz in this experiment. The position of the randomly selected solution is the variable sPD which is equal to s . The randomized cumulative effect of all prairie dogs in the colony is denoted by the term, $DPD_{x,y}$, which is determined in equation (21). The strength of the coterie in digging defined as CDS is a random value determined by the quality of a food source as given in the equation (22). Lastly the Levy distribution which is known to facilitate superior and efficient search space exploration.

$$fDbest_{x,y} = Hbest_{x,y} \times \Delta + \frac{PD_{x,y} \times \text{mean}(PD_{o,m})}{Hbest_{x,y} \times (UPB_y - LWB_y) \Delta} \quad (22)$$

$$DPD_{x,y} = \frac{Hbest_{x,y} - sPD_{x,y}}{Hbest_{x,y} + \Delta} \quad (23)$$

$$CDS = 1.5 \times s \times \left(1 - \frac{\text{iter}}{\text{Max}_{\text{iter}}} \right)^{\left(2 \frac{\text{iter}}{\text{Max}_{\text{iter}}} \right)} \quad (24)$$

The stochasticity introduced by the variable s is necessary so that exploration can occur, this variable takes the value of either -1 or 1 which is alternated with the value of the other solution depending on whether the current iteration is an odd or even number. The constant parameter small constant, denoted as D , has been used to explain the disparities between particular prairie dogs, but the PDO implementation uses prairie dogs of identical nature. In this case, the maximum number of iterations is denoted by Max_{iter} and the current iteration number is denoted by iter . This section presents the exploitative section of the PD algorithm. This stage is carried out using two strategies that are described in equation (25) and (26). As indicated above the PD alternates between these two strategies according to the current iteration

$\frac{\text{Max}_{\text{iter}}}{2} \leq \text{iter} < 3 \frac{\text{Max}_{\text{iter}}}{4}$ and $3 \frac{\text{Max}_{\text{iter}}}{4} \leq \text{iter} \leq \text{Max}_{\text{iter}}$ respectively.

$$PD_{x+1,y+1} = Hbest_{x,y} - fDbest_{x,y} \times \varepsilon - DPD_{x,y} \times \text{rand} \forall \frac{\text{Max}_{\text{iter}}}{2} \leq \text{iter} < 3 \frac{\text{Max}_{\text{iter}}}{4} \quad (25)$$

$$PD_{x+1,y+1} = Hbest_{x,y} \times QE \times \text{rand} \forall 3 \frac{\text{Max}_{\text{iter}}}{4} \leq \text{iter} < \text{Max}_{\text{iter}} \quad (26)$$

Let $Hbest_{x,y}$ be the best solution for the global solution achieved. The global, $fDbest_{x,y}$ is the assessment of the impact of the best solution in the present. The value of ε is a little constant that depicts the quality of food source. Equation (19) also defines the cumulative impact of all the prairie dogs in the colony $DPD_{x,y}$. Also where QE is the predator effect, as defined in equation (27) and random number taken over the range of (0, 1).

$$QE = 1.5 \times \left(1 - \frac{\text{iter}}{\text{Max}_{\text{iter}}} \right)^{\left(2 \frac{\text{iter}}{\text{Max}_{\text{iter}}} \right)} \quad (27)$$

Here iter denotes the current iteration, and Max_{iter} represents the maximum number of iterations. Algorithm for prairie dog optimization is illustrated in Table 2.

Table 2: Algorithm for prairie dog optimization

Fixed the PDOA parameters o, n, ρ, ε Fixed $Hbest$ and $Dbest$ as ϕ Modify the candidate solution DS and PD While $\text{iter} < \text{Max}_{\text{iter}}$ do For ($x = 1$ to n) do For ($y = 1$ to o) do Analyse the fitness of PD
--

```

Discovery the best solution
Modernise  $Hbest$ 
Modernise  $CDS$  and  $QE$  using equation (20) and (26)
Modernise  $DPD_{x,y}$  using equation (19)
If  $\left( iter < \frac{Max_{iter}}{4} \right)$  then
     $PD_{x+1,y+1} = Hbest_{x,y} - eDbest_{x,y} \times \rho - DPD_{x,y} \times Levy(o)$ 
Else if  $\left( \frac{Max_{iter}}{4} \leq iter < 3 \frac{Max_{iter}}{2} \right)$  then
     $PD_{x+1,y+1} = Hbest_{x,y} - eDbest_{x,y} \times CDS \times Levy(o)$ 
Else if  $\left( \frac{Max_{iter}}{2} \leq iter < 3 \frac{Max_{iter}}{4} \right)$ 
     $PD_{x+1,y+1} = Hbest_{x,y} - eDbest_{x,y} \times \varepsilon - DPD_{x,y} \times rand$ 
Else
     $PD_{x+1,y+1} = Hbest_{x,y} \times QE \times rand$ 
End if
End for
End for
 $iter = iter + 1$ 
End while
Return
End

```

A meta-heuristic technique that is little yet strong enough to balance exploration and expropriation by mimicking PD social foraging and burrow-building activities. Its adaptive motion systems avoid the early convergence and maintain a great worldwide search capacity, therefore making it very suited for solving complex optimisation issues. Through the proper tuning of hyper parameters, PDOA improves model accuracy, convergence speed, and provides resilience against a range of problem areas.

3.4 FedAvg

In the federated learning (FL) model, the object is to reduce a global objective functional, which is a weighted average of loss functionalities of all the personal datasets of the clients. Assume that we have a client population of size Z , with each client, z holding a private dataset E_z . The global objective can be stated to be.

$$\min_a g(a) := \sum_{z=1}^Z \frac{o_z}{o} G_z(a) \quad (28)$$

In this case the parameters of the global model that should be trained are denoted by a . The notation o_z denotes the size of the local dataset of client E_z , and o . The local objective function G_z usually corresponds to a loss function like the cross-entropy loss in the supervised classification operation. A client-server model-based FedAvg is a synchronous federated learning algorithm that serves as a baseline [25]. Each round the server sends the global model to a random group of clients, who are training locally and training it on a few epochs. The clients send the difference between their models and the received models back to the server. These updates are summed by the server based on the size of the datasets of the clients and applied to the global model through SGD with the learning rate of 1. In a case of hardware-limited devices, there is a risk that some clients fail to meet the update deadline, leading to issues of fairness. Asynchronous versions of FedAvg solve this by permitting updates to occur at other times. But in this work, the conventional synchronous FedAvg baseline is considered.

4. Results and discussion

The analysis was carried out with the Python programming language on a computer with an Intel(R) Core(TM) i7-4770S processor operating at 3.40 GHz. The machine had 16.00 GB of RAM (15.9 GB available) and an operating system that was 64-bit.

4.1 Dataset Description

NIH Chest X-rays Dataset: Some of the simplest and least costly medical tests to carry out are chest X-ray tests. Nonetheless, reading of X-rays of the chest may be more difficult compared to chest computers tomography. The biggest challenge when constructing such datasets is the scarcity of resources to label large quantities of images. The largest open source before this dataset, OpenI, had 4,143 chest X-ray images. The NIH Chest X-ray Dataset has 112,120 images/pictures of 30,805 distinct patients whose disease information has been labeled. The labels were created through Natural Language Processing application in deriving disease classifications of associated radiology reports [26]. They have been estimated to be more than 90 percent accuracy and weakly-supervised learning. Though the initial reports are not publicly accessible, the description of the labeling procedure can be obtained in the open-access article: ChestX-ray8: Hospital-scale Chest X-ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases. The dataset is normally divided 2 types 80% of training and 20% of testing respectively.

4.2 Performance Evaluation

The performance of the proposed model is analysed by using the performance metrics such as accuracy, precision, recall, F1-score, computation time, latency, and response time. Analysis of accuracy of proposed and existing models is evaluated in Figure 3.

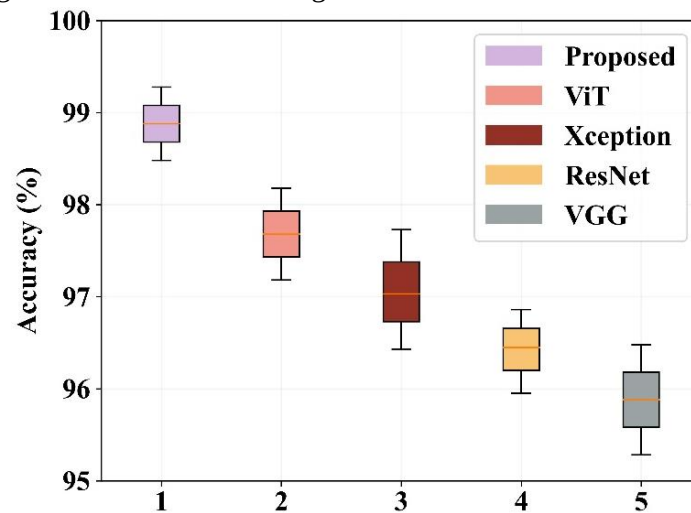


Figure 3: Analysis of accuracy for proposed and existing models

The OSDSTran model attained meaningfully much higher performance compared to other existing models such as VGG [27], ResNet [28], Xception [29], and ViT [30] which reached accuracies of 95.88%, 96.45%, 97.03%, and 97.68% respectively. In contrast, the proposed model obtained an impressive accuracy of 98.88% it will highlighting its effectiveness in demand prediction. This will also reduce the false negatives. Analysis of precision for proposed and existing models is illustrated in Figure 4.

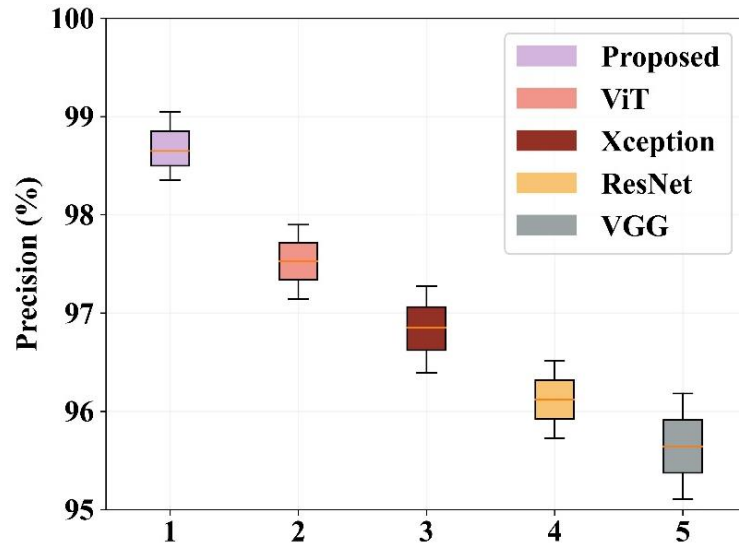


Figure 4: Analysis of precision for proposed and existing models

The figure illustrate the precision of the OSDSTran model attained better than other existing methods. The other models are VGG, ResNet, Xception, and ViT their respective precision rates of 95.64%, 96.12%, 96.85% and 97.53%. Also the proposed OSDSTran model attained a better high precision score of 98.65%. Which significant improvement the proposed model capture more complex features and improve the classification accuracy. Comparison of recall for OSDSTran model and existing models is present in Figure 5.

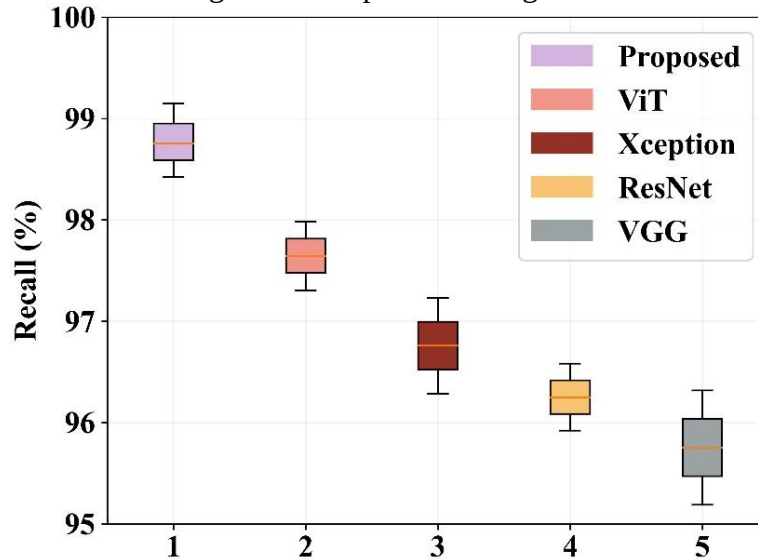


Figure 5: Comparison of recall for OSDSTran model and existing models

The superior performance in recall is shown by the OSDSTran model which scores an amazing 98.75%, considerably outperforming the scores of VGG at 95.75%, ResNet at 96.25%, Xception at 96.75% and ViT at 97.64%. The remarkable higher recall value is accurately identify the models true negatives and false negatives. Analysis of F1-score for proposed model and existing models is show in Figure 6.

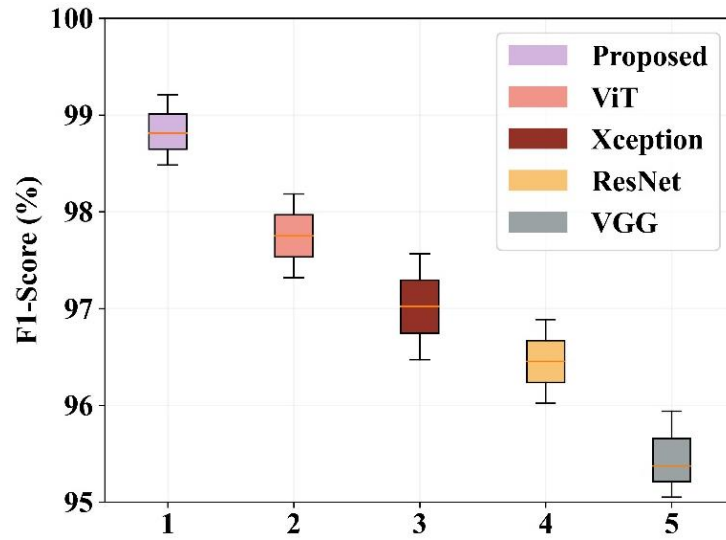


Figure 6: Analysis of F1-score for proposed model and existing models

The graph describes the F1-score for OSDSTran model and several existing models with traditional models such VGG, ResNet, Xception, and ViT scoring 95.37%, 96.45%, 97.02%, and 97.75%, correspondingly. Conversely the suggested OSDSTran model achieved an outstanding F1-score of 98.81%. The high F1-score indicate OSDSTran model well balance in both precision and recall which make system more efficiently. Compression of average computation time for OSDSTran model and existing model is given in Figure 7.

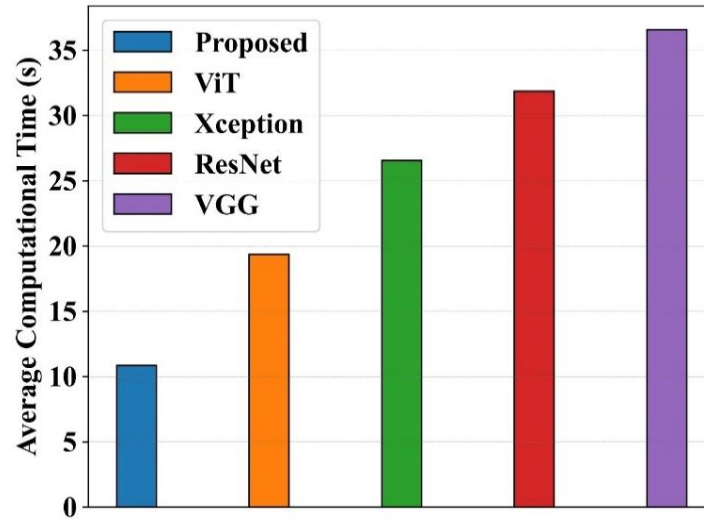


Figure 7: Compression of average computation time for OSDSTran model and existing model

The average computation time of the proposed model is far higher than that of several existing models, such as VGG, ResNet, Xception, and ViT with 36.58 s, 31.85 s, 26.54 s and 19.35 s accuracy values respectively. However the new model ensured an average computation time of 10.85 s. This low average computation time for OSDSTran model which have attained high performance and low computational expensive in real-world processing. Analysis of latency for OSDSTran model and existing model is illustrated in Figure 8.

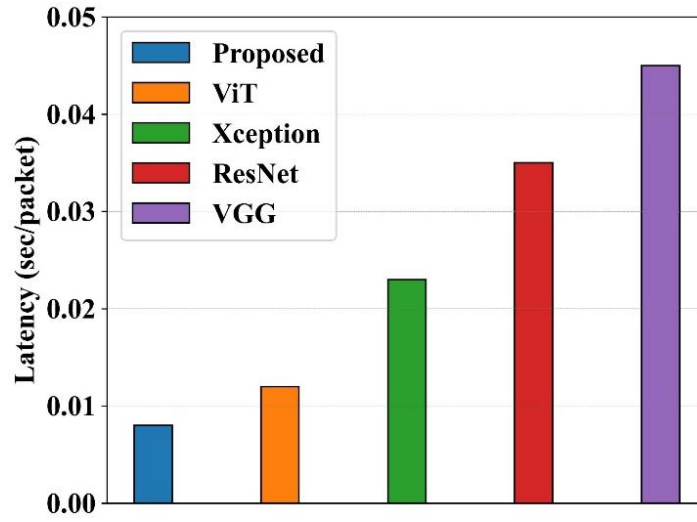


Figure 8: Analysis of latency for OSDSTran model and existing model

The latency of the proposed model considerably exceeds those in many previously constructed models. One such existing models include VGG, ResNet, Xception, and ViT that produced values for 0.045 sec/packet, 0.035 sec/packet, 0.023 sec/packet and 0.012 sec/packet respectively but a new model that emerged perfectly was achieved with a low value of 0.008 sec/packet. Analysis for response time for proposed model is present in Figure 9.

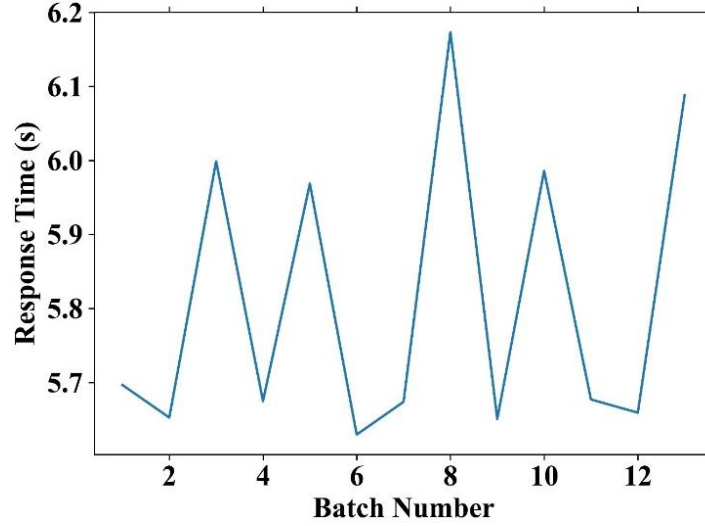


Figure 9: Analysis for response time for proposed model

The response time graph of FL model indicates variation among the various batches, where time varies between 5.5 and 6.1 seconds. Nevertheless, the model has a constant processing time, with periodic spikes, which corresponds to the possible effect of other factors, such as the complexity of the data, delays in communication between local and global systems, or changing computational loads of different batches. Nonetheless, the range of response is stable, and all the variations do not exceed a controllable range, indicating the efficiency of the model in processing IoV data to identify network intrusion. AUC for proposed model is illustrated in Figure 10.

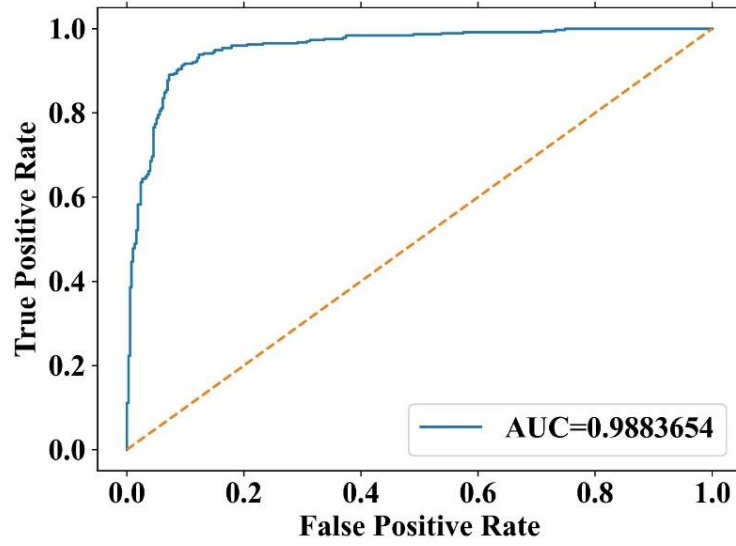


Figure 10: AUC for proposed model

OSDSTran model registers high AUC values of 0.9883 respectively. In addition to demonstrating the models impressive classification accuracy, high AUC highlights its potential applications in industries that require precision in their decisions and risk assessment. It is also a viable real-time processing tool because it enhances the performance of the system. PRC for OSDSTran model is illustrated in Figure 11.

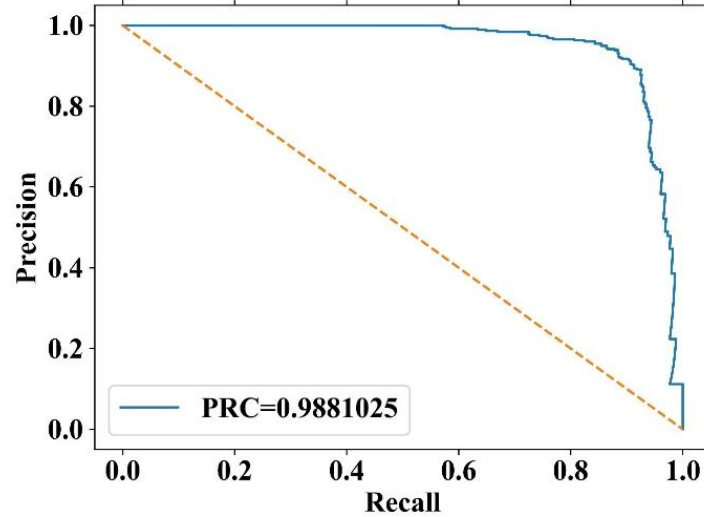


Figure 11: PRC for OSDSTran model

The OSDSTran model is recorded high PRC values of 0.9881 respectively. Beyond showing the models remarkable classification accuracy, this high PRC emphasizes its possible for practical uses in sectors needing exact decision-making and risk analysis. It is also a useful tool for real-time processing as it improves system performance. Analysis of training, testing loss and accuracy of the proposed model is represented in the Figure 12.

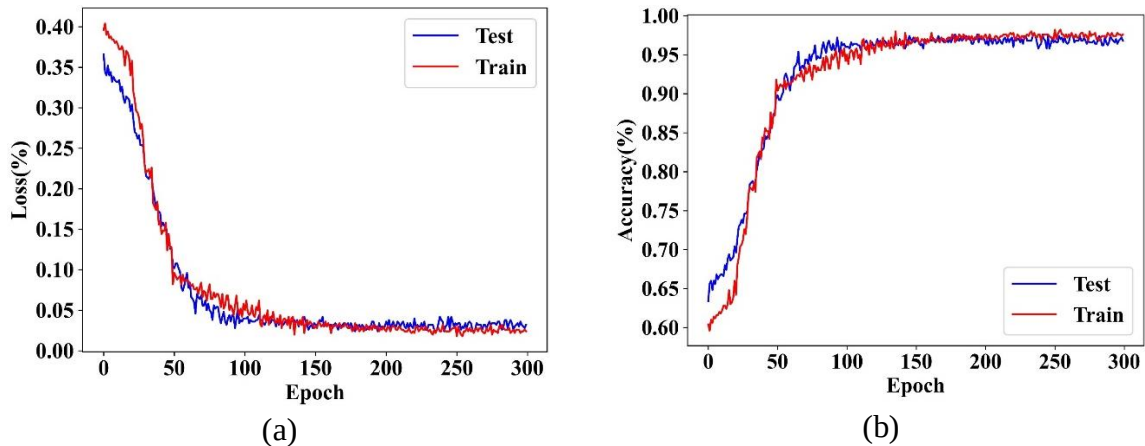


Figure 12: Analysis of training, testing loss and accuracy of the proposed model

The proposed model, training, testing loss, and accuracy of analysis. The models had varying accuracy and loss epoch values at varying training value points. Further, the healthcare device authentication system is analysed using training and testing models. Accuracy and loss analysis of epoch value up-and-down format-based analysis models. Maximum accuracy of the training and testing output would result in improved performance. At varying points in testing values, each model had different accuracy and loss epoch values. This approach model involves testing the dataset through the testing model. Accuracy of testing and loss epoch value up-and-down format-based analysis models. Maximum training and test outcome accuracy results in an improved performance.

4.3 Discussion

The existing models faced several disadvantages such as Ahmed et al. [16] possessed suffer reduced operational time due to the frequent authentication. In paper proposed model utilize HMAC-SHA256 for authentication with low computational which also reduce operational time. Additionally Ayyappan et al. [17] had different hospital nodes or IoT devices may requires high training. The proposed model used PDOA for hyper-parameter tuning which will improve the classification accuracy and reduce the training time. Moreover Yang et al. [18] model required high computational power in training time this may increase computational cost. Proposed model used optimized model for classification which will minimize the computational power. Furthermore Eshwarappa et al. [19] the model dependence on image quality was affected in reliability of real-time clinical use. Proposed model used Swin transformer approach efficiently captures the local and global features making it highly effective for accurate diseases classification. Finally Chen et al. [20] complexity of the model was limited generalization issues. OSDSTran is utilized PDOA which improve model generalization. The analysis of the proposed model is illustrated in table 4.

Table 4: Comparative analysis of the proposed model

Author name & References	Models	Accuracy (%)
Ahmed et al. [16]	Medical Edge-devices privacy enhanced technique based on the federated learning	96
Ayyappan et al. [17]	Federated edge AI	91.2
Yang et al. [18]	Clu-FDL	93
Eshwarappa et al. [19]	Edge-Avg	93.94
Chen et al. [20]	AQ-FDNF	93.8
Proposed	OSDSTran	98.88

5. Conclusion

A FL framework has been presented in this paper that provides privacy to IoMT systems where the most challenging aspects of privacy include data confidentiality, secure model aggregation, and strong authentication. As opposed to a conventional FL, which is susceptible to malicious updates, the offered scheme proposes a two-layered authentication scheme, which is backed by a CAA. The interactive model training process can be involved only to trusted hospitals and devices and prevent potential threats of non-verified and corrupted nodes since HMAC-SHA256 authentication and active token validation should be applied. This framework requires an OSDSTran to classify local medical image to ensure high diagnosis and privacy. The mechanism of spatial attention of the network can be simply trained to notice the significant parts of the chest X-ray images and to memorize the local and global interdependencies to identify the disease in the right way. Additionally, the PDOA is presented to optimise the hyper-parameters, and this matter is relevant in improving model convergence and classification. The proposed solution will make sure that any sensitive patient data will be highly localized in all

hospital location and will only share and aggregate updates of models across FedAvg. The experimental assessment the proposed model attained the accuracy of 98.88% and precision of 98.65%. In the future work will implement the Blockchain-based method is used to improve the transparency and verification.

References

31. Khan, Habib Ullah, Yasir Ali, and Faheem Khan. "A features-based privacy preserving assessment model for authentication of internet of medical things (IoMT) devices in healthcare." *Mathematics* 11, no. 5 (2023): 1197.
32. Das, Sangjukta, and Suyel Namasudra. "Lightweight and efficient privacy-preserving mutual authentication scheme to secure internet of things-based smart healthcare." *Transactions on Emerging Telecommunications Technologies* 34, no. 11 (2023): e4716.
33. Ryu, Hyunho, and Hyunsung Kim. "Privacy-preserving authentication protocol for wireless body area networks in healthcare applications." In *Healthcare*, vol. 9, no. 9, p. 1114. MDPI, 2021.
34. Azad, Muhammad Ajmal, Junaid Arshad, Shazia Mahmoud, Khaled Salah, and Muhammad Imran. "A privacy-preserving framework for smart context-aware healthcare applications." *Transactions on Emerging Telecommunications Technologies* 33, no. 8 (2022): e3634.
35. Soni, Mukesh, and Dileep Kumar Singh. "Privacy-preserving authentication and key-management protocol for health information systems." In *Data Protection and Privacy in Healthcare*, pp. 37-50. CRC Press, 2021.
36. Rajasekaran, Arun Sekar, Azees Maria, Maheswar Rajagopal, and Josip Lorincz. "Blockchain enabled anonymous privacy-preserving authentication scheme for internet of health things." *Sensors* 23, no. 1 (2022): 240.
37. Almaiah, Mohammed Amin, Fahima Hajjej, Aitizaz Ali, Muhammad Fermi Pasha, and Omar Almomani. "A novel hybrid trustworthy decentralized authentication and data preservation model for digital healthcare IoT based CPS." *Sensors* 22, no. 4 (2022): 1448.
38. Kathamuthu, Nirmala Devi, Annadurai Chinnamuthu, Nelson Iruthayanathan, Manikandan Ramachandran, and Amir H. Gandomi. "Deep Q-learning-based neural network with privacy preservation method for secure data transmission in internet of things (IoT) healthcare application." *Electronics* 11, no. 1 (2022): 157.
39. Kute, Shruti, A. V. Shreyas Madhav, Amit Kumar Tyagi, and Atharva Deshmukh. "Authentication Framework for Healthcare Devices Through Internet of Things and Machine Learning." In *Evolutionary Computing and Mobile Sustainable Networks: Proceedings of ICECMSN 2021*, pp. 399-415. Singapore: Springer Singapore, 2022.
40. Singh, Saurabh, Shailendra Rathore, Osama Alfarraj, Amr Tolba, and Byungun Yoon. "A framework for privacy-preservation of IoT healthcare data using Federated Learning and blockchain technology." *Future Generation Computer Systems* 129 (2022): 380-388.
41. Ali, Mansoor, Faisal Naeem, Muhammad Tariq, and Georges Kaddoum. "Federated learning for privacy preservation in smart healthcare systems: A comprehensive survey." *IEEE journal of biomedical and health informatics* 27, no. 2 (2022): 778-789.
42. Sabry, Farida, Tamer Eltaras, Wadha Labda, Khawla Alzoubi, and Qutaibah Malluhi. "Machine learning for healthcare wearable devices: the big picture." *Journal of Healthcare Engineering* 2022, no. 1 (2022): 4653923.
43. Li, Xiong, Jiabei He, Pandi Vijayakumar, Xiaosong Zhang, and Victor Chang. "A verifiable privacy-preserving machine learning prediction scheme for edge-enhanced HCPSs." *IEEE Transactions on Industrial Informatics* 18, no. 8 (2021): 5494-5503.
44. Soni, Mukesh, and Dileep Kumar Singh. "Privacy-preserving authentication and key-management protocol for health information systems." In *Data Protection and Privacy in Healthcare*, pp. 37-50. CRC Press, 2021.
45. Unal, Devrim, Shada Bennbaia, and Ferhat Ozgur Catak. "Machine learning for the security of healthcare systems based on Internet of Things and edge computing." In *Cybersecurity and cognitive science*, pp. 299-320. Academic Press, 2022.
46. Ahmed, Syed Thouheed, A. C. Kaladevi, Achyut Shankar, and Fayez Alqahtani. "Privacy enhanced edge-ai healthcare devices authentication: a federated learning approach." *IEEE*

Transactions on Consumer Electronics (2025).

47. Ayyappan, G., V. Loganathan, E. Padma, S. Ilavarasan, and A. Subash. "Federated Learning and Edge AI for Privacy-Preserving Diabetes Prediction in Healthcare." In 2025 3rd International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS), pp. 1116-1121. IEEE, 2025.
48. Yang, Xinyi, and Juan Li. "A clustering-based federated deep learning approach for enhancing diabetes management with privacy-preserving edge artificial intelligence." *Healthcare Analytics* 7 (2025): 100392.
49. Eshwarappa, Nithin Melala, Hojjat Baghban, Ching-Hsien Hsu, Po-Yen Hsu, Ren-Hung Hwang, and Mu-Yen Chen. "Communication-efficient and privacy-preserving federated learning for medical image classification in multi-institutional edge computing." *Journal of Cloud Computing* 14, no. 1 (2025): 44.
50. Chen, Ke, and Long Xiao. "Energy-efficient privacy-preserving AI models for real-time health monitoring in mobile IoT networks for professional sports applications." *Discover Applied Sciences* 7, no. 9 (2025): 1-25.
51. Choi, Hojin, and Seog Chung Seo. "Optimization of PBKDF2 using HMAC-SHA2 and HMAC-LSH families in CPU environment." *IEEE Access* 9 (2021): 40165-40177.
52. Lin, Ailiang, Bingzhi Chen, Jiayu Xu, Zheng Zhang, Guangming Lu, and David Zhang. "Ds-transunet: Dual swin transformer u-net for medical image segmentation." *IEEE Transactions on Instrumentation and Measurement* 71 (2022): 1-15.
53. Zhang, Xiangrong, Shouwang Shang, Xu Tang, Jie Feng, and Licheng Jiao. "Spectral partitioning residual network with spatial attention mechanism for hyperspectral image classification." *IEEE transactions on geoscience and remote sensing* 60 (2021): 1-14.
54. Biswas, Saptadeep, Azharuddin Shaikh, Absalom El-Shamir Ezugwu, Japie Greeff, Seyedali Mirjalili, Uttam Kumar Bera, and Laith Abualigah. "Enhanced prairie dog optimization with Levy flight and dynamic opposition-based learning for global optimization and engineering design problems." *Neural Computing and Applications* 36, no. 19 (2024): 11137-11170.
55. Zhou, Yuhao, Qing Ye, and Jiancheng Lv. "Communication-efficient federated learning with compensated overlap-fedavg." *IEEE Transactions on Parallel and Distributed Systems* 33, no. 1 (2021): 192-205.
56. <https://www.kaggle.com/datasets/nih-chest-xrays/data>
57. Kareem, Amer, Haiming Liu, and Vladan Velisavljevic. "A privacy-preserving approach to effectively utilize distributed data for malaria image detection." *Bioengineering* 11, no. 4 (2024): 340.
58. Adnan, Amna, Firdous Kausar, Muhammad Shoaib, Faiza Iqbal, Ayesha Altaf, and Hafiz M. Asif. "A Secure and Privacy-Preserving Approach to Healthcare Data Collaboration." *Symmetry* 17, no. 7 (2025): 1139.
59. Patel, Nisarg P., Raj Parekh, Saad Ali Amin, Rajesh Gupta, Sudeep Tanwar, Neeraj Kumar, Rahat Iqbal, and Ravi Sharma. "LEAF: A federated learning-aware privacy-preserving framework for healthcare ecosystem." *IEEE Transactions on Network and Service Management* 21, no. 1 (2023): 1129-1141.
60. Amin, Al, Kamrul Hasan, Sharif Ullah, and M. Shamim Hossain. "ViT Enhanced Privacy-Preserving Secure Medical Data Sharing and Classification." In 2025 IEEE 22nd Consumer Communications & Networking Conference (CCNC), pp. 1-2. IEEE, 2025.



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